

# **MATH1141: Higher Mathematics 1A (Calculus)**

L. Cheung

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# Chapter 1

## Sets, inequalities and functions

### 1.1 Sets of numbers

A set is a collection of distinct objects. The objects in a set are elements or members of the set. Commonly used sets:

- The set of natural numbers:  $\mathbb{N} = \{0, 1, 2, 3, 4, \dots\}$
- The set of integers:  $\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3, \dots\}$
- The set of rational numbers:  $\mathbb{Q} = \{\frac{p}{q} : p, q \text{ are integers and } q \neq 0\}$
- The set of  $\mathbb{R}$  real numbers may be represented as the collection of points lying on the number line

If  $A$  is a set of numbers and the number  $x$  is a member of the set  $A$ , then we write:

$$x \in A$$

If  $x$  is not a member of  $A$ , then we write

$$x \notin A$$

#### 1.1.1 Intervals

- Parenthesis: excludes endpoints
- Bracket: includes endpoints
- Combination: neither open nor closed interval

#### 1.1.2 Solving Inequalities

For  $x, y, z \in \mathbb{R}$ :

- if  $x > y$ , then  $x + z > y + z$
- if  $x > y$  and  $z > 0$ , then  $xz > yz$
- if  $x > y$  and  $z < 0$ , then  $xz < yz$

### Example

Solve the quadratic inequality  $x^2 + 4x > 21$

$$\begin{aligned}
 x^2 + 4x > 21 &\iff x^2 + 4x - 21 > 0 \\
 &\iff (x + 7)(x - 3) > 0
 \end{aligned}$$

Solution set:  $x < -7$  or  $x > 3$

$$x \in (-\infty, -7) \cup (3, \infty)$$

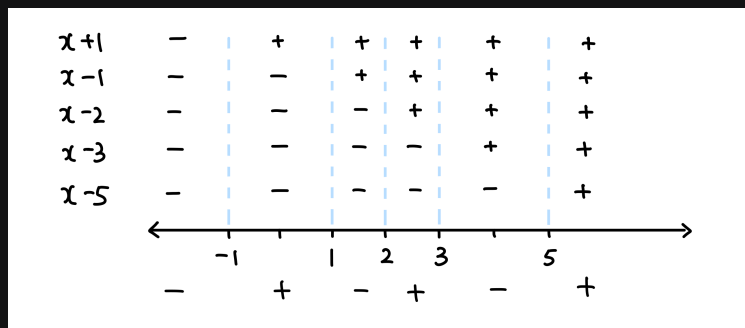
### Example

Solve the rational inequality  $\frac{1}{x+1} < \frac{1}{(x-2)(x-3)}$

Multiply both sides by  $(x+1)^2(x-2)^2(x-3)^2$  ( $>= 0$ ),

$$\begin{aligned}
 \frac{1}{x+1} < \frac{1}{(x-2)(x-3)} &\iff (x+1)(x-2)^2(x-3)^2 < (x+1)^2(x-2)(x-3) \\
 &\iff (x+1)(x-2)^2(x-3)^2 - (x+1)^2(x-2)(x-3) < 0 \\
 &\iff (x+1)(x-2)(x-3)(x-4)(x-5) < 0
 \end{aligned}$$

Solution set:  $(-\infty, -1) \cup (1, 2) \cup (3, 5)$



## 1.1.3 Absolute Values

The absolute value (or magnitude) of a real number  $x$  is

$$|x| = \begin{cases} x & \text{if } x \geq 0, \\ -x & \text{if } x < 0 \end{cases}$$

### Example

Solve the inequality  $|3x+1| \leq 4$

$$\begin{aligned}
 |3x+1| \leq 4 &\iff \left|x + \frac{1}{3}\right| \leq \frac{4}{3} \\
 &\iff x + \frac{1}{3} \leq \frac{-4}{3} \quad \text{or} \quad x + \frac{1}{3} \leq \frac{4}{3} \\
 &\iff x \leq \frac{-5}{3} \quad \text{or} \quad x \leq 1
 \end{aligned}$$

Solution set:  $(-\infty, \frac{-5}{3}] \cup [1, \infty)$

#### Example

Solve the inequality  $\frac{|x+5|}{|x-11|} < 1$

$$\frac{|x+5|}{|x-11|} < 1 \iff |x+5| < |x-11| \quad (\text{multiply both sides by } |x-11|)$$

$$\iff |x+5|^2 < |x-11|^2$$

$$\iff x^2 + 10x + 25 < x^2 - 22x + 121$$

$$\iff 32x < 96$$

$$\iff x < 3$$

Solution set:  $(-\infty, 3)$

**Note:** 11 is not in the solution set

### 1.1.4 Functions

#### Domain, codomain and range

Let  $A$  and  $B$  be subsets of  $\mathbb{R}$ : A function with domain  $A$  and codomain  $B$  is a rule which assigns to every  $x \in A$  exactly one number  $f(x) \in B$

$$f : A \rightarrow B$$

$$f : A \ni x \mapsto f(x) \in B$$

- $A = \text{Dom}(f)$ : set of all inputs
- $B = \text{Codom}(f)$ : set that contains all the outputs
- Range of  $f$ : set of actual outputs, defined by:

$$\text{Range}(f) = \{f(x) \in B : x \in A\}$$

- Hence,  $\text{Range}(f) \subseteq \text{Codom}(f)$
- The **codomain does not have to be unique**
- The codomain is useful when the range is unknown

#### Example

Given:  $f : [0, \infty) \rightarrow \mathbb{R}, f(x) = 2 + \sqrt{x}, \forall x \in [0, \infty)$

Then,  $\text{Dom}(f) = [0, \infty)$ ,  $\text{Codom}(f) = \mathbb{R}$ , and  $\text{Range}(f) = [2, \infty)$

Natural domain (or maximal domain) of  $f$  is the largest possible domain for which the rule makes sense (for real numbers)

#### Forming new functions

Let  $A, B, C$  and  $D$  be subsets of  $\mathbb{R}$ . Given two functions  $f : A \rightarrow B$  and  $g : A \rightarrow B$ , then the functions

$$f + g, \quad f - g, \quad fg, \quad \frac{f}{g}$$

are defined by the rules

$$(f + g)(x) = f(x) + g(x), \quad \forall x \in A$$

$$(f - g)(x) = f(x) - g(x), \quad \forall x \in A$$

$$(fg)(x) = f(x)g(x), \quad \forall x \in A$$

$$\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)}, \quad \forall x \in A \text{ provided that } g(x) \neq 0.$$

To form these new functions, the domains of both  $f$  and  $g$  must be the same

Suppose that  $f : C \rightarrow D$  and  $g : A \rightarrow B$  are functions such that

$$\text{Range}(g) \subseteq \text{Dom}(f) = C$$

Then the composition function  $(f \circ g)(x) = f(g(x)), \quad \forall x \in A$

### 1.1.5 The Elementary Functions

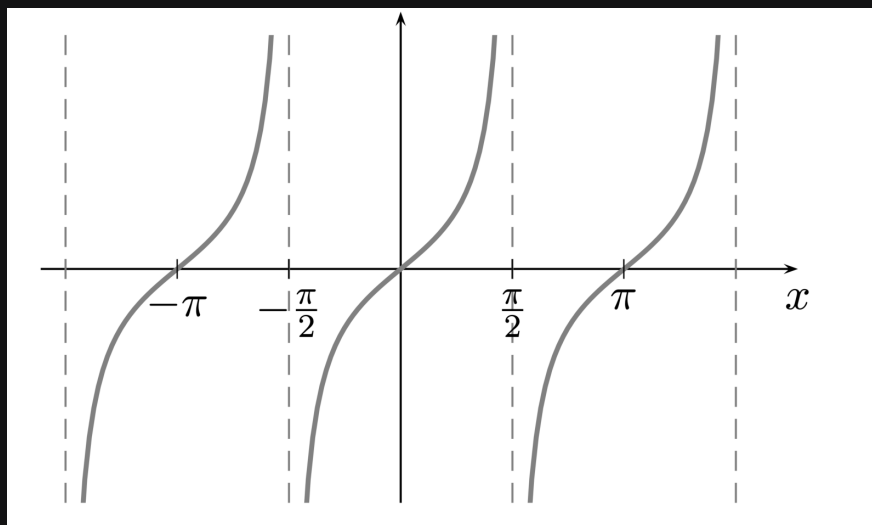
Elementary functions are those that can be constructed by combining a finite number of polynomials, exponentials, logarithms, roots, trigonometric and inverse trigonometric functions via function composition ( $\circ$ ),  $+$ ,  $-$ ,  $\times$ , and  $\div$ .

### 1.1.6 Implicitly Defined Functions

Many curves on the plane can be described by the points  $(x, y)$  that satisfy some equation involving  $x$  and  $y$ . However, not all these curves are necessarily functions. Some cannot be expressed with a single  $y$  term ( $y = \dots$ ). In this case, the curve may be decomposed into two or more functions that are implicitly defined by the curve.

### 1.1.7 Continuous Functions

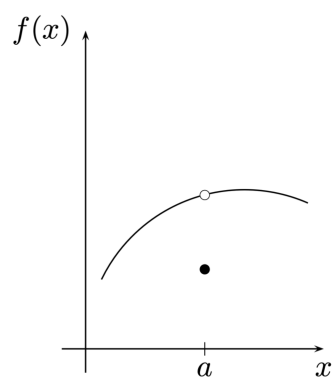
In the below example, the  $\tan x$  graph breaks at  $x = -\frac{\pi}{2}, \frac{\pi}{2}, \dots$



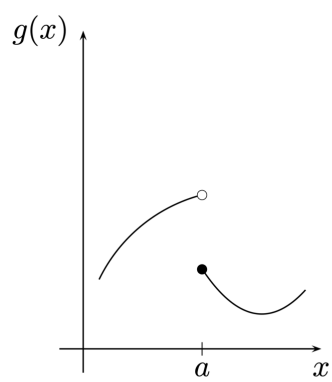
However, this break in the domain does not make it a discontinuity in the graph. Discontinuities can be categorised as following:



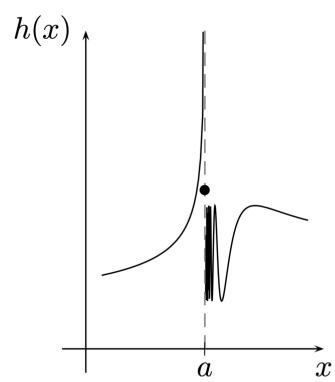
removable



jump



essential



# Chapter 2

## Limits

### 2.1 Limits of functions at infinity

- If  $f(x)$  gets arbitrarily close to some  $L \in \mathbb{R}$  as  $x$  tends to  $\infty$ , then we write:

$$\lim_{x \rightarrow \infty} f(x) = L \quad \text{or} \quad f(x) \rightarrow L \text{ as } x \rightarrow \infty$$

- If  $f(x)$  gets arbitrarily large as  $x$  tends to  $\infty$ , then we write:

$$f(x) \rightarrow \infty \text{ as } x \rightarrow \infty$$

- In this case, we state that the limit of  $f(x)$  as  $x$  tends to  $\infty$  does not exist

$$\lim_{x \rightarrow \infty} \frac{1}{f(x)} = 0$$

- The limit notation can only be used when the limit has a constant definition (do not write  $\lim_{x \rightarrow \infty} = \infty$ )

#### 2.1.1 Basic rules for limits

If  $\lim_{x \rightarrow \infty} f(x) = L_1$  and  $\lim_{x \rightarrow \infty} g(x) = L_2$  then

- $\lim_{x \rightarrow \infty} (f(x) + g(x)) = L_1 + L_2$
- $\lim_{x \rightarrow \infty} (f(x) - g(x)) = L_1 - L_2$
- $\lim_{x \rightarrow \infty} f(x)g(x) = L_1L_2$
- $\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = \frac{L_1}{L_2}$ , provided that  $L_2 \neq 0$

### Example

Find the limit of  $f(x) = \frac{1 + \frac{2}{3x+4}}{5 - 6e^{-x}}$  as  $x \rightarrow \infty$  (if it exists)

$$\begin{aligned}\lim_{x \rightarrow \infty} f(x) &= \frac{\lim_{x \rightarrow \infty} (1 + \frac{2}{3x+4})}{\lim_{x \rightarrow \infty} (5 - 6e^{-x})} \\&= \frac{\lim_{x \rightarrow \infty} 1 + \lim_{x \rightarrow \infty} \frac{2}{3x+4}}{\lim_{x \rightarrow \infty} 5 - \lim_{x \rightarrow \infty} 6e^{-x}} \\&= \frac{1+0}{5-0} \\&= \frac{1}{5}\end{aligned}$$

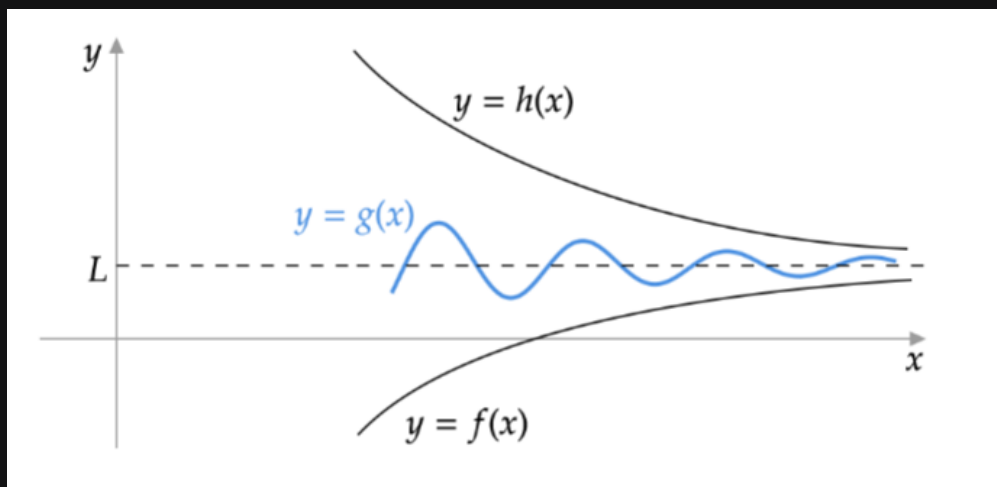
### 2.1.2 The pinching theorem

Suppose that  $f$  and  $h$  have the same limit as  $x \rightarrow \infty$ . If the graph of  $g$  always lies between the graphs of  $f$  and  $h$ , then  $g$  has the same limit.

**Theorem** Let  $f$ ,  $g$ , and  $h$  be functions defined on  $(b, \infty)$  for some  $b \in \mathbb{R}$ . If :

- $f(x) \leq g(x) \leq h(x)$ ,  $\forall x \in (b, \infty)$ , and
- $\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} h(x) = L$ , then

$$\lim_{x \rightarrow \infty} g(x) = L$$



### Example

Find the limit of  $g(x) = e^{-x^2} 2^{\sin 3x}$  as  $x \rightarrow \infty$  (if it exists).

$$\sin x \in [-1, 1], \quad \forall x \in \mathbb{R}$$

$$\Rightarrow \frac{1}{2} \leq 2^{\sin 3x} \leq 2, \quad \forall x \in \mathbb{R}$$

$$\Rightarrow \frac{e^{-x^2}}{2} \leq e^{x^2} 2^{\sin 3x} \leq 2e^{-x^2}$$

$$\text{Since } \lim_{x \rightarrow \infty} \frac{e^{-x^2}}{2} = \lim_{x \rightarrow \infty} 2e^{-x^2} = 0,$$

$$\therefore \lim_{x \rightarrow \infty} e^{-x^2} 2^{\sin 3x} = 0$$

### 2.1.3 Limits of the form $f(x)/g(x)$

When calculating  $\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)}$  when both  $f(x)$  and  $g(x)$  tend to  $\infty$  as  $x \rightarrow \infty$ , the previous rules cannot be applied. The main goal in this situation is to **divide both  $f$  and  $g$  by the fastest growing term in  $g$** .

### Example

Find the limit  $\lim_{x \rightarrow \infty} \frac{5x^3 + 6x^2 - 4 \sin x}{\cos 3x + 5x - 2x^3}$  (if it exists).

$$\begin{aligned} \frac{5x^3 + 6x^2 - 4 \sin x}{\cos 3x + 5x - 2x^3} &= \frac{5 + \frac{6}{x} - 4 \frac{\sin x}{x^3}}{\frac{\cos 3x}{x^3} + \frac{5}{x^2} - 2} \\ &= \frac{5 + 0 + 0}{0 + 0 - 2} \\ &= -\frac{5}{2} \quad \text{as } x \rightarrow \infty \end{aligned}$$

### Example

Find the limit  $\lim_{x \rightarrow \infty} \frac{x^2 + 3x}{\sqrt{2x^4 + 3} - 4x}$  (if it exists).

$$\begin{aligned} \frac{x^2 + 3x}{\sqrt{2x^4 + 3} - 4x} &= \frac{1 + \frac{3}{x}}{\frac{1}{x^2} \sqrt{2x^4 + 3} - \frac{4}{x}} \\ &= \frac{1 + \frac{3}{x}}{\sqrt{2 + \frac{3}{x^4}} - \frac{4}{x}} \\ &\rightarrow \frac{1 + 0}{\sqrt{2 + 0} - 0} \\ &= \frac{1}{\sqrt{2}} \quad \text{as } x \rightarrow \infty \end{aligned}$$

### 2.1.4 Limits of the form $\sqrt{f(x)} - \sqrt{g(x)}$

In this case, the leading term may not help. Instead the lower order term will usually determine the limit. This can be achieved by **multiplying the numerator and denominator by the conjugate**.

#### Example

Find the limit of  $\lim_{x \rightarrow \infty} \sqrt{2x^2 + 3x} - \sqrt{2x^2 - x}$ .

$$\begin{aligned} \sqrt{2x^2 + 3x} - \sqrt{2x^2 - x} &= \frac{(\sqrt{2x^2 + 3x} - \sqrt{2x^2 - x})(\sqrt{2x^2 + 3x} + \sqrt{2x^2 - x})}{(\sqrt{2x^2 + 3x} + \sqrt{2x^2 - x})} \\ &= \frac{2x^2 + 3x - (2x^2 - x)}{\sqrt{2x^2 + 3x} + \sqrt{2x^2 - x}} \\ &= \frac{4x}{\sqrt{2x^2 + 3x} + \sqrt{2x^2 - x}} \\ &= \frac{x}{\sqrt{2 + \frac{3}{x}} + \sqrt{2 - \frac{1}{x}}} \\ &\Rightarrow \frac{4}{\sqrt{2} + \sqrt{2}} \\ &= \sqrt{2} \quad \text{as } x \rightarrow \infty \end{aligned}$$

### 2.1.5 Indeterminate forms

Some limits such as the form  $\frac{\infty}{\infty}$  cannot be determined in advance and is hence called an indeterminate form. Other indeterminate forms include:

- $\infty - \infty$  if  $f(x) \rightarrow \infty$  and  $g(x) \rightarrow \infty$  as  $x \rightarrow \infty$ ,  $f(x) - g(x) \rightarrow ?$
- $\frac{0}{0}$  if  $f(x) \rightarrow 0$  and  $g(x) \rightarrow 0$  as  $x \rightarrow \infty$ ,  $\frac{f(x)}{g(x)} \rightarrow ?$
- $0 \times \infty$  if  $f(x) \rightarrow 0$  and  $g(x) \rightarrow \infty$  as  $x \rightarrow \infty$ ,  $f(x)g(x) \rightarrow ?$

## 2.2 The definition of $\lim_{x \rightarrow \infty} f(x)$

Suppose that  $b, L \in \mathbb{R}$ , and  $f$  is a real-valued function defined on  $(b, \infty)$ . It is said:

$$\lim_{x \rightarrow \infty} f(x) = L$$

if for every  $\epsilon > 0$ , there is  $M \in \mathbb{R}$  such that:

$$x > M \Rightarrow |f(x) - L| < \epsilon$$

## 2.3 Proving that $\lim_{x \rightarrow \infty} f(x) = L$ using the limit definition

### Example

Prove that  $\lim_{x \rightarrow \infty} \frac{5x}{x+3} = 3$ .

$|f(x) - L| < \epsilon$  if?

$$\begin{aligned} |f(x) - L| &= \left| \frac{5x}{x+3} - 3 \right| = \left| \frac{5x - 3(x+3)}{x+3} \right| = \left| \frac{-15}{x+3} \right| \\ &= \frac{15}{x+3} \quad \text{for } x > 3 \\ &< \frac{15}{x} \quad \text{for } x > 0 \end{aligned}$$

$$\Rightarrow |f(x) - L| < \epsilon \quad \text{if} \quad \frac{15}{x} < \epsilon \text{ and } x > 0$$

$$\text{(i.e. } x > \frac{15}{\epsilon} > 0)$$

$$\Rightarrow \text{For any } \epsilon > 0, \exists M = \frac{15}{\epsilon} \text{ such that}$$

$$|f(x) - L| < \epsilon \text{ whenever } x > M$$

### Strategy for finding $M$

1. Find a good upper bound for  $|f(x) - L|$
2. Find a simple condition on  $x$  such that this upper bound is less than  $\epsilon$
3. Use this condition to state an appropriate value for  $M$

**Note:** The value of  $M$  is not unique. If  $M_0$  is a value for  $M$ , then any upper bound of  $M_0$  is also a value for  $M$

### Example

Show that  $\lim_{x \rightarrow \infty} \frac{x^2 - \cos x}{x^2 + 1} = 1$

$$\begin{aligned} \left| \frac{x^2 - \cos x}{x^2 + 1} - 1 \right| &= \left| \frac{x^2 - \cos x - (x^2 + 1)}{x^2 + 1} \right| = \left| \frac{-\cos x - 1}{x^2 + 1} \right| = \left| \frac{\cos x + 1}{x^2 + 1} \right| \\ &\leq \frac{2}{x^2 + 1} \\ &\leq \frac{2}{x^2} \end{aligned}$$

$$\Rightarrow |f(x) - 1| < \epsilon \quad \text{if} \quad \frac{2}{x^2} < \epsilon$$

$$\therefore \frac{2}{x^2} < \epsilon \Leftrightarrow \sqrt{\frac{2}{\epsilon}} < x \quad \text{for } x > 0, \quad M \triangleq \sqrt{\frac{2}{\epsilon}}$$

## 2.4 Proofs of basic limit results

Not assessable.

## 2.5 Limits of functions at a point

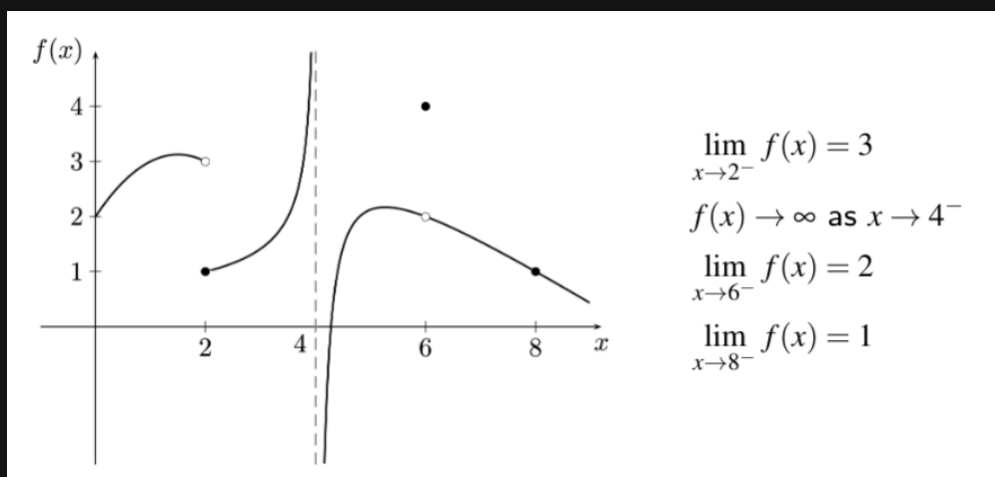
### 2.5.1 Left-hand, right-hand and two-sided limits

- If  $f(x)$  approaches  $L$  as  $x$  approaches  $a$  from the left-hand side, then:

$$\lim_{x \rightarrow a^-} f(x) = L \quad (\text{left-hand limit})$$

- If  $f(x)$  gets arbitrarily large as  $x$  approaches  $a$  from the left side, then:

$$f(x) \rightarrow \infty \quad \text{as } x \rightarrow a^-$$

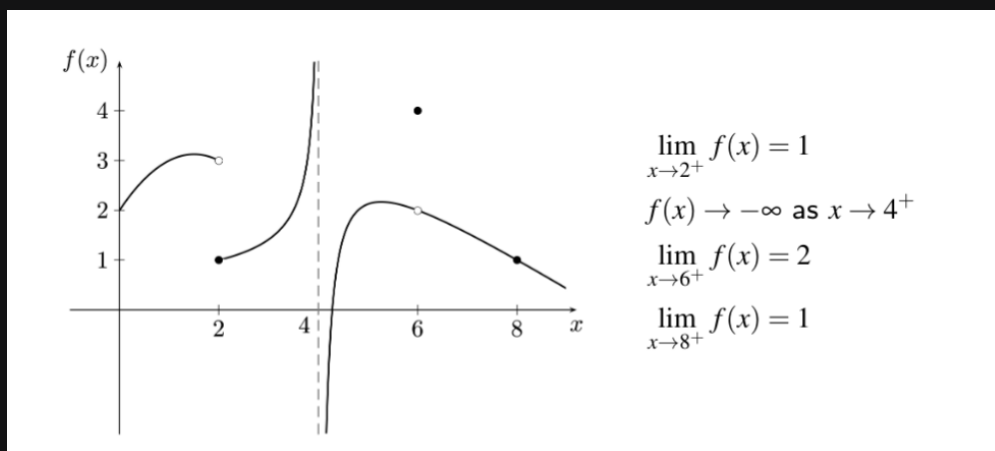


- If  $f(x)$  approaches  $L$  as  $x$  approaches  $a$  from the right-hand side, then:

$$\lim_{x \rightarrow a^+} f(x) = L \quad (\text{right-hand limit})$$

- If  $f(x)$  gets arbitrarily large as  $x$  approaches  $a$  from the right side, then:

$$f(x) \rightarrow \infty \quad \text{as } x \rightarrow a^+$$



- Left  $I$  be an open interval that contains the point  $a$ . For a real-valued function  $f$  defined on  $I$  except possibly at  $a$ , if both one-sided limits exist and

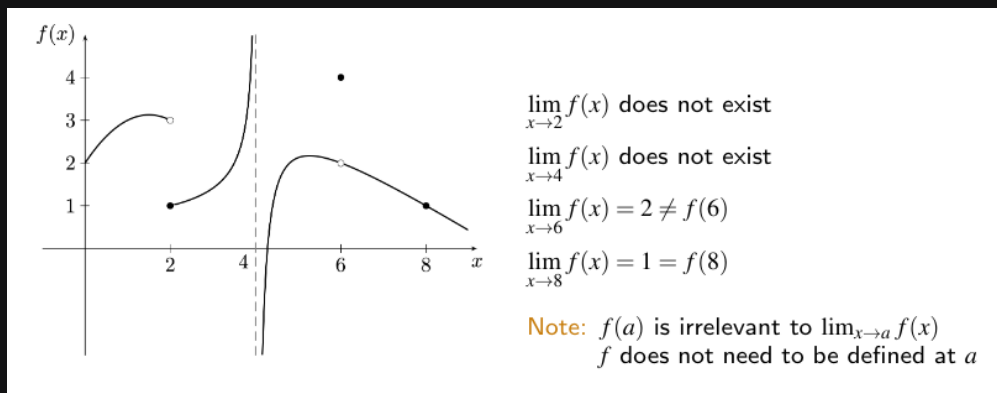
$$\lim_{x \rightarrow a^-} f(x) = \lim_{x \rightarrow a^+} f(x) = L,$$

then we say that the (two-sided) limit of  $f(x)$  as  $x \rightarrow a$  exists and is equal to  $L$ , and we write

$$\lim_{x \rightarrow a} f(x) = L.$$

Otherwise,  $\lim_{x \rightarrow a} f(x)$  does not exist.

- Note:**  $f(a)$  is irrelevant to  $\lim_{x \rightarrow a} f(x)$ .  $f$  does not need to be defined at  $a$



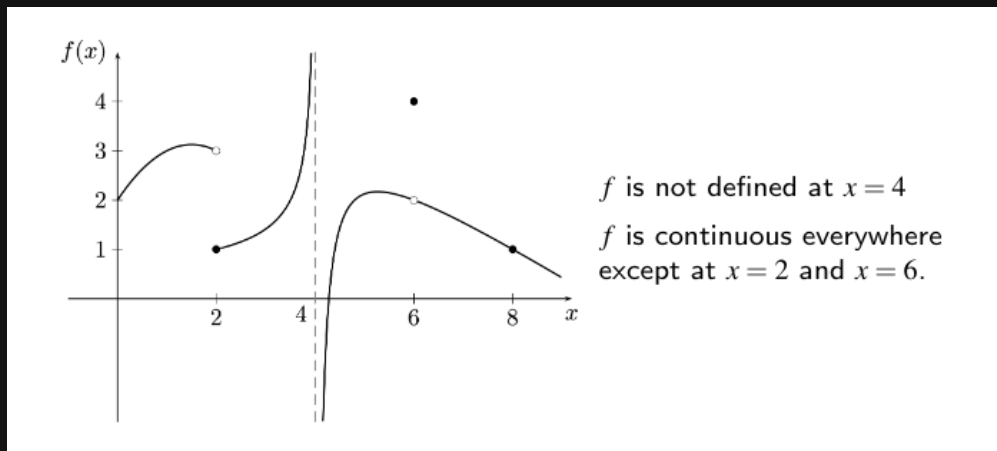
## 2.5.2 Limits and continuous functions

Let  $I$  be an open interval that contains the point  $a$ . A function  $f : I \rightarrow \mathbb{R}$  is continuous at  $a$  if

$$\lim_{x \rightarrow a} f(x) = f(a)$$

Otherwise, we say that  $f$  is discontinuous at  $a$ .

If  $f$  is continuous at every point of  $\text{Dom}(f)$ , then we say that  $f$  is continuous everywhere.



## 2.5.3 Rules for limits at a point

### Arithmetic

Suppose that  $a \in \mathbb{R}$ , and  $\lim_{x \rightarrow a} f(x) = L_1$  and  $\lim_{x \rightarrow a} g(x) = L_2$ . Then



- $\lim_{x \rightarrow a} (f(x) + g(x)) = L_1 + L_2$
- $\lim_{x \rightarrow a} (f(x) - g(x)) = L_1 - L_2$
- $\lim_{x \rightarrow a} f(x)g(x) = L_1 L_2$
- $\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{L_1}{L_2}$

### Function composition

If  $\lim_{x \rightarrow a} f(x) = L$  and  $g$  is continuous at  $L$ , then

$$\lim_{x \rightarrow a} g(f(x)) = g\left(\lim_{x \rightarrow a} f(x)\right) = g(L)$$

#### Example

Find the limit  $\lim_{x \rightarrow 5} \sqrt{2x + \cos \pi x^2}$ .

$$\begin{aligned} \lim_{x \rightarrow 5} \sqrt{2x + \cos \pi x^2} &= \sqrt{\lim_{x \rightarrow 5} 2x + \cos \pi x^2} \quad (\sqrt{\phantom{x}} \text{ is continuous}) \\ &= \sqrt{\lim_{x \rightarrow 5} 2x + \lim_{x \rightarrow 5} \cos \pi x^2} \\ &= \sqrt{\lim_{x \rightarrow 5} 2x + \cos \lim_{x \rightarrow 5} \pi x^2} \quad (\cos \text{ is continuous}) \\ &= \sqrt{10 + \cos 5^2 \pi} \quad (\cos 5^2 \pi = -1) \\ &= 3 \end{aligned}$$

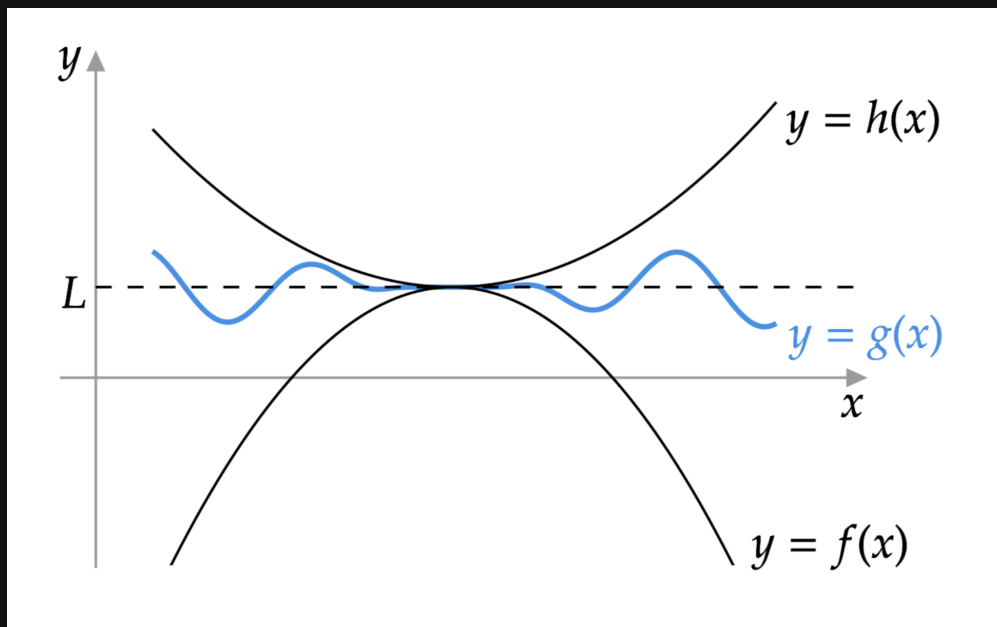
### Pinching theorem (two-sided limit)

Let  $I$  be an open interval that contains the point  $a$ . Suppose that  $f$ ,  $g$ , and  $h$  are functions defined on  $I$ . If

- $f(x) \leq g(x) \leq h(x), \quad \forall x \in I$ , and
- $\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} h(x) = L$

then

$$\lim_{x \rightarrow a} g(x) = L$$



#### Example

Suppose that  $f : \mathbb{R} \rightarrow \mathbb{R}$  is defined by the formula

$$f(x) = \begin{cases} \frac{|x^2-9|}{x-3} & \text{if } x \neq 3 \\ -6 & \text{if } x = 3 \end{cases}$$

Discuss the limiting behaviour of  $f(x)$  as  $x$  approaches 3.

Limit from left hand side

$$\begin{aligned} \text{For } x \in (-3, 3), \quad \frac{|x^2-9|}{x-3} &= \frac{|x+3||x-3|}{x-3} = -(x+3) \\ \Rightarrow \lim_{x \rightarrow 3^-} f(x) &= \lim_{x \rightarrow 3^-} -(x+3) = -6 = f(3) \end{aligned}$$

Limit from right hand side

$$\begin{aligned} \text{For } x > 3, \quad \frac{|x^2-9|}{x-3} &= x+3 \\ \Rightarrow \lim_{x \rightarrow 3^+} f(x) &= \lim_{x \rightarrow 3^+} x+3 = 6 \end{aligned}$$

Since  $\lim_{x \rightarrow 3^+} f(x) \neq \lim_{x \rightarrow 3^-} f(x)$ , two-sided limit does not exist at  $x = 3$

So,  $f$  is not continuous at  $x = 3$

## Chapter 3

# Properties of continuous functions

### 3.1 Combining continuous functions

Suppose that the functions  $f$  and  $g$  are continuous at a point  $a$ . Then

1.  $f + g$ ,  $f - g$  and  $fg$  are continuous at  $a$
2.  $\frac{f}{g}$  is continuous at  $a$ , provided that  $g(a) \neq 0$

$$\begin{aligned}\lim_{x \rightarrow a} (f + g)(x) &= \lim_{x \rightarrow a} (f(x) + g(x)) && \text{(by definition of } f + g) \\ &= \lim_{x \rightarrow a} f(x) + \lim_{x \rightarrow a} g(x) && \text{(by "limit of sum = sum of limits")} \\ &= f(a) + g(a) && \text{(by continuity of } f \text{ and } g \text{ at } a) \\ &= (f + g)(a)\end{aligned}$$

Suppose that the function  $f$  is continuous at  $a$ , and  $g$  is continuous at  $f(a)$ . Then  $g \circ f$  is continuous at  $a$ .

$$\begin{aligned}\lim_{x \rightarrow a} (g \circ f)(x) &= \lim_{x \rightarrow a} g(f(x)) \\ &= g\left(\lim_{x \rightarrow a} f(x)\right) && \text{(by limit rule of function composition)} \\ &= g(f(a)) && \text{(by continuity of } f \text{ at } a) \\ &= (g \circ f)(a)\end{aligned}$$

#### Example

Every polynomial  $p : \mathbb{R} \rightarrow \mathbb{R}$  is continuous at every  $a \in \mathbb{R}$ .

$$\begin{aligned}\Leftarrow \text{for any } n \in \mathbb{N}, \quad p(x) &= a_n x^n + a_{n-1} x^{n-1} + \cdots + a_0 \\ &= \text{product \& sum of } g(x) = x \text{ and constant functions, ie. continuous by (i)}\end{aligned}$$

### 3.2 Continuity on intervals

For a real-valued function  $f$  defined on an open interval  $(a, b)$ , we say that  $f$  is continuous on  $(a, b)$  if it is continuous at every  $x \in (a, b)$ .

For a real-valued function  $f$  defined on a closed interval  $[a, b]$ , we say that

- $f$  is continuous at the endpoint  $a$  if  $\lim_{x \rightarrow a^+} f(x) = f(a)$
- $f$  is continuous at the endpoint  $b$  if  $\lim_{x \rightarrow b^-} f(x) = f(b)$
- $f$  is continuous on  $[a, b]$  if it is continuous on  $(a, b)$  and at endpoints  $a$  and  $b$ .

### 3.3 The intermediate value theorem

Suppose that  $f$  is continuous on  $[a, b]$ , and  $f(b) < z < f(a)$  for some  $z$ . The graph of  $f$  from  $(a, f(a))$  to  $(b, f(b))$  without lifting the pen. We must cross a point  $(c, z)$  along the way for some  $c \in (a, b)$ .

Let  $f : [a, b] \rightarrow \mathbb{R}$  be a continuous function. If either  $f(a) < z < f(b)$  or  $f(b) < z < f(a)$  for some  $z \in \mathbb{R}$ , then there is at least one  $c \in (a, b)$  such that  $f(c) = z$

- In general,  $c$  is not unique
- The continuity of  $f$  is necessary
- Only for continuous functions defined on (subsets of)  $\mathbb{R}$ , no analogue for  $\mathbb{Q}$

$$\begin{array}{ll} f : \mathbb{R} \rightarrow \mathbb{R}, & f(x) = x^2 - 2 \\ & f \text{ is continuous on } \text{Dom}(f) = \mathbb{R} \\ & = \in (f(0), f(2)), f(c) = 0 \text{ at } c = \sqrt{2} \in (0, 2) \\ f : \mathbb{Q} \rightarrow \mathbb{Q}, & g(x) = x^2 - 2 \\ & g \text{ is continuous on } \text{Dom}(g) = \mathbb{Q} \\ & = \in (g(0), g(2)), g(c) = 0 \text{ at } c = \sqrt{2} \notin \mathbb{Q} \end{array}$$

- Since infinity is not part of  $\mathbb{R}$ , a definite bound must be defined before using IVT

#### Example

Show that there exists a solution  $c \in [1, 2]$  of the equation  $\sqrt{c} = c^2 - 1$  and approximate its value.

Define a function  $f : [1, 2] \rightarrow \mathbb{R}$  by  $f(x) = x^2 - 1 - \sqrt{x}$ . Then,

- $f$  is continuous on  $[1, 2]$
- $f(1) = -1$  and  $f(2) = 3 - \sqrt{2} > 0$ .

Since  $f(1) < 0 < f(2)$ , by **IVT**, there is a number  $c \in (1, 2)$  such that  $f(c) = c^2 - 1 - \sqrt{c} = 0$

Approximation:

- take  $c_1 = 1.5 \Rightarrow f(c_1) = 0.0253$ , by IVT,  $\exists$  a solution  $c \in (1, c_1)$
- take  $c_2 = 1.25 \Rightarrow f(c_2) = -0.5555$ , by IVT,  $\exists$  a solution  $c \in (1, c_2)$

### 3.4 The maximum-minimum theorem

Let  $f$  be a real-valued function defined on a closed interval  $[a, b]$ .

- A point  $c \in [a, b]$  is a global minimum point for  $f$  on  $[a, b]$  if

$$f(c) \leq f(x), \quad \forall x \in [a, b].$$

- A point  $d \in [a, b]$  is a global maximum point for  $f$  on  $[a, b]$  if

$$f(d) \geq f(x), \quad \forall x \in [a, b].$$

If  $f$  is continuous on  $[a, b]$ , then  $f$  attains its minimum and maximum on  $[a, b]$ , ie. there exists points  $c, d \in [a, b]$  such that

$$f(c) \leq f(x) \leq f(d), \quad \forall x \in [a, b].$$

- Closed intervals are necessary
- Continuity of  $f$  is necessary

#### Example

Let  $f$  be a function defined by  $f(x) = \begin{cases} \frac{1}{x^2+1} \sin \frac{1}{x}, & \text{if } x \neq 0 \\ 1, & \text{if } x = 0 \end{cases}$

1. Is  $f$  bounded on  $[-2, 2]$ ?

$$\text{Yes. For } x \neq 0, \quad |f(x)| = \left| \frac{1}{x^2+1} \sin \frac{1}{x} \right| \leq \frac{1}{x^2} \leq 1.$$

$$\text{For } x = 0, \quad f(x) = 1.$$

$$\Rightarrow |f(x)| \leq 1, \quad \forall x \in [-2, 2]$$

2. Does it attain a minimum or maximum on  $[-2, 2]$

- $f$  on  $[-2, 2]$  attains a maximum at  $x = 0$ .
- $f$  does not have a minimum value on  $[-2, 2]$

$$\text{For } x_n = \left( \frac{-\pi}{2} + 2n\pi \right)^{-1}, \quad n \in \mathbb{Z}$$

$$\sin \frac{1}{x_n} = -1$$

$$f(x_n) = -\frac{1}{\left( \frac{-\pi}{2} + 2n\pi \right)^{-2} + 1}$$

Then as  $n \rightarrow \infty$ ,  $x_n \rightarrow 0$  and  $f(x_n) \rightarrow -1$

Note:  $f(x) > -1$

$\Rightarrow$  as  $x \rightarrow 0$ ,  $f(x)$  oscillates between (not touching)  $-1$  and  $1$

## Chapter 4

# Differentiable functions

### 4.1 Gradients of tangents and derivatives

Suppose that  $f$  is defined on some open interval containing the point  $x$ . It is said that  $f$  is differentiable at  $x$  if

$$\lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \text{ exists}$$

This limit is denoted by  $f'(x)$ ,  $\frac{df}{dx}(x)$  or  $\frac{d}{dx}f(x)$  and is called the derivative of  $f$  at  $x$ .

### Example

1. Show that  $\lim_{h \rightarrow 0} \frac{\cos(h) - 1}{h} = 0$ . (Hint:  $\cos(h) - 1 = \frac{\cos^2(h) - 1}{\cos(h) + 1} = -\frac{\sin^2(h)}{\cos(h) + 1}$ )

$$\begin{aligned} \lim_{h \rightarrow 0} \frac{\cos(h) - 1}{h} &= -\lim_{h \rightarrow 0} \frac{\sin(h)}{\cos(h) + 1} \frac{\sin(h)}{h} \\ &= -\frac{\lim_{h \rightarrow 0} \sin(h)}{\lim_{h \rightarrow 0} \cos(h) + 1} \lim_{h \rightarrow 0} \frac{\sin(h)}{h} \quad \left( \lim_{h \rightarrow 0} \frac{\sin(h)}{h} = 1 \right) \\ &= -\frac{0}{2} \times 1 = 0 \end{aligned}$$

2. For which  $n \in \mathbb{N}$  is the function  $f$  defined by

$$f(x) = \begin{cases} x^n \sin\left(\frac{1}{x}\right) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

continuous and differentiable at  $x = 0$

For continuity,  $\lim_{x \rightarrow 0} x^n \sin\left(\frac{1}{x}\right) = f(0) = 0$ .

$$1 \leq \sin\left(\frac{1}{x}\right) \leq 1, \forall x \neq 0 \quad \Rightarrow \quad -|x^n| \leq x^n \sin\left(\frac{1}{x}\right) \leq |x^n|, \forall x \neq 0, \forall n \in \mathbb{N}$$

$$(\text{Pinching theorem}) \quad \Rightarrow \quad \lim_{x \rightarrow 0} x^n \sin\left(\frac{1}{x}\right) = 0, \quad \text{ie. continuous at } x = 0, \forall n \in \mathbb{N} \setminus \{0\}$$

For differentiability,  $\lim_{h \rightarrow 0} \frac{1}{h} [f(0+h) - f(0)]$  exists.

$$\begin{aligned} &= \frac{1}{h} \left[ h^n \sin\left(\frac{1}{h}\right) - 0 \right] \\ &= h^{n-1} \sin\left(\frac{1}{h}\right) \rightarrow 0, \quad \forall n-1 \in \mathbb{N} \setminus \{0\} \end{aligned}$$

ie.  $n \in \mathbb{N}, n \neq 0, 1$

## 4.2 Rules for differentiation

Suppose that  $f$  and  $g$  are differentiable at  $x$ . Then  $f + g$ ,  $f - g$  and  $fg$  are differentiable at  $x$ , and  $\frac{f}{g}$  is also differentiable at  $x$ , provided  $g(x) \neq 0$ .

1.  $(f + g)'(x) = f'(x) + g'(x)$
2.  $(Cf)'(x) = Cf'(x)$  where  $C$  is any constant
3.  $(fg)'(x) = f'(x)g(x) + f(x)g'(x)$ , (product rule)
4.  $\left(\frac{f}{g}\right)'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{(g(x))^2}$ , provided that  $g(x) \neq 0$  (quotient rule)

Suppose that  $g$  is differentiable at  $x$ , and  $f$  is differentiable at the point  $g(x)$ . Then  $f \circ g$  is differentiable at  $x$  and

$$(f \circ g)'(x) = f'(g(x))g'(x)$$

### 4.3 Proofs of results

lol no

### 4.4 Implicit differentiation

A function defined by an equation relating  $x$  and  $y$  is implicitly defined.

#### Example

Determine the tangent to the curve  $x^4 - x^2y^2 + y^4 = 13$  at  $(2, 1)$

$y$  depends on  $x$ :  $y(x)$

in a neighbourhood of  $(2, 1)$

Differentiate both sides of the equation with respect to  $x$ :

$$4x^3 - 2xy^2(x) - 2x^2y(x)y'(x) + 4y^3(x)y'(x) = 0$$

At  $(x, y) = (2, 1)$ ,

$$32 - 4 - 8y'(2) + 4y'(2) = 0$$

$$28 - 4y'(2) = 0$$

$$y'(2) = 7$$

Tangent line at  $(2, 1)$ :

$$y - 1 = 7(x - 2)$$

### 4.5 Differentiation, continuity and split functions

- If  $f$  is differentiable at  $a$ , then  $f$  is continuous at  $a$
- If  $f$  is not continuous at  $a$ , then it is not differentiable at  $a$
- If  $f$  is continuous at  $a$ , it may not be differentiable at  $a$

– eg.  $f: \mathbb{R} \rightarrow \mathbb{R}$ ,  $f(x) = |x|$  and  $a = 0$

Suppose  $a \in \mathbb{R}$  and the function  $f$  is defined by

$$f(x) = \begin{cases} p(x) & \text{if } x \geq a \\ q(x) & \text{if } x < a \end{cases}$$

where  $p$  and  $q$  are differentiable in some open interval containing  $a$ . If

- $f$  is continuous at  $a$  ( $\Leftrightarrow p(a) = q(a)$ ) and
- $p'(a) = q'(a)$ ,

then  $f$  is differentiable at  $x = a$ .



## 4.6 Derivatives and function approximation

If  $f$  is differentiable at  $a$ , then

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

By the definition of a limit,  $\frac{f(x) - f(a)}{x - a}$  gets close to  $f'(a)$  as  $x$  gets close to  $a$ . This statement is equivalent to

$$\begin{aligned}\frac{f(x) - f(a)}{x - a} &\approx f'(a) \\ f(x) &\approx f(a) + f'(a)(x - a)\end{aligned}$$

### Example

Estimate  $\sin(0.51\pi)$

$$\begin{aligned}\sin(0.51\pi) &\approx \sin(0.5\pi) + \cos(0.5\pi)(0.51\pi - 0.5\pi) \\ &= 1 + 0 \times (0.01\pi) \\ &= 1 \\ \Rightarrow \sin(0.51\pi) &\approx 1\end{aligned}$$

## 4.7 Derivatives and rates of change

ext 1 content

## 4.8 Local maximum, local minimum and stationary points

Let  $f$  be defined on some interval  $I$ .

- A point  $c \in I$  is a local minimum if there exists a  $h > 0$  such that

$$f(x) \geq f(c), x \in (c - h, c + h) \cap I$$

- A point  $d \in I$  is a local maximum point if there exists a  $h > 0$  such that

$$f(x) \leq f(d), x \in (d - h, d + h) \cap I$$

## Chapter 5

# The mean value theorem and applications

### 5.1 The mean value theorem

Suppose that  $f$  is

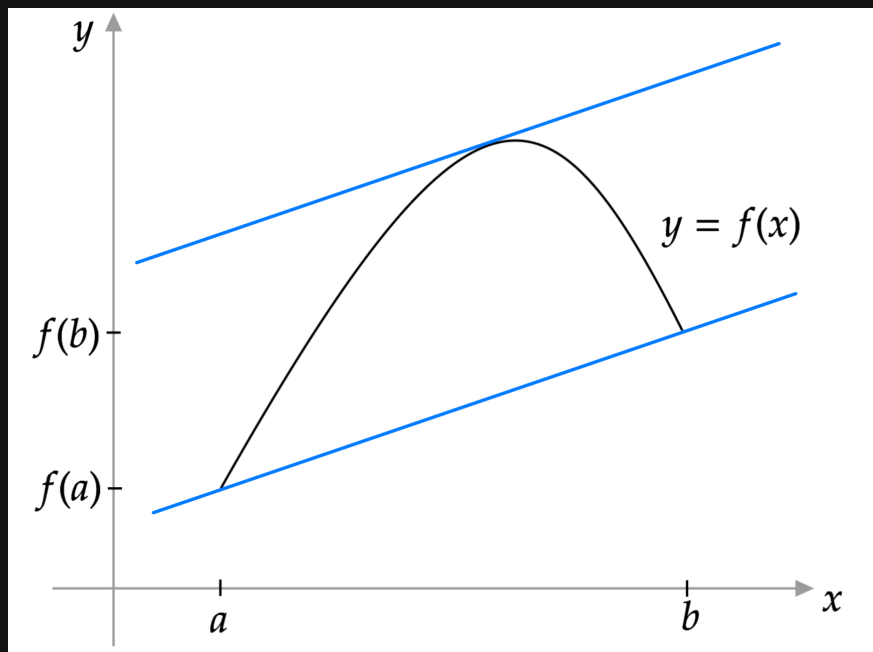
1. continuous on  $[a, b]$ , and
2. differentiable on  $(a, b)$

Then there is at least one  $c \in (a, b)$  such that

$$\frac{f(b) - f(a)}{b - a} = f'(c)$$

The mean value theorem cannot be applied if

- $f$  is not continuous on  $[a, b]$
- $f$  is not differentiable on  $(a, b)$



#### Example

A car enters a tunnel at a speed of 30km/h and after 1 minute leaves the tunnel at a speed of 40km/h. The length of the tunnel is 1km. Did the driver break the speed limit of 50km/h?

Let  $x : [t_{in}, t_{out}] \rightarrow \mathbb{R}$ ,  $x(t)$  = position of the car at time  $t$

Assume  $x$  is continuous on  $[t_{in}, t_{out}]$  and differentiable on  $(t_{in}, t_{out})$

$\Rightarrow$  By MVT,  $\exists T \in (t_{in}, t_{out})$  such that

$$x'(t) = \frac{x(t_{out}) - x(t_{in})}{t_{out} - t_{in}} = \frac{1}{\frac{1}{60}} \text{ km/h} = 60 \text{ km/h}$$

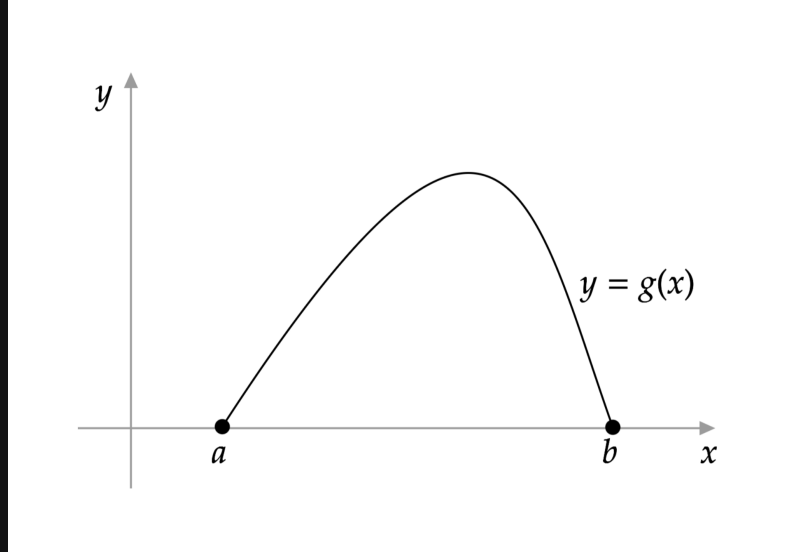
Therefore, the driver broke the speed limit

## 5.2 Proof of the mean value theorem

Suppose that  $g$

- is continuous on  $[a, b]$
- differentiable on  $(a, b)$
- $g(a) = g(b) = 0$  (optional)

Then there is  $c \in (a, b)$  such that  $g'(c) = 0$



**Case 1:**  $g(x) = 0, \forall x \in [a, b]$   
 $\Rightarrow g'(c) = 0$  for any  $c \in (a, b)$

**Case 2:**  $g(d) > 0$  for some  $d \in (a, b)$   
 By continuity and MMT,  $g$  attains a maximum value at some  $c \in [a, b]$

Then  $g(c) \geq g(d) > 0 = g(a) = g(b)$   
 $\Rightarrow c \neq a, c \neq b$ , so  $c \in (a, b)$   
 $\Rightarrow g'(c) = 0$

**Case 3:**  $g(x) \leq 0, \forall x \in [a, b]$ , and  $g$  is not a constant on  $[a, b]$   
 Since  $g(a) = g(b)$ , Case 3  $\Leftrightarrow g(d) < 0$  for some  $d \in (a, b)$   
 Similar to Case 2,  $g$  attains a minimum value at some  $c \in (a, b)$ ,  $g'(c) = 0$

To prove MVT, a function  $f$  is "deformed" such that it looks like  $g$  in Rolle's theorem.

Define  $g : [a, b] \rightarrow \mathbb{R}$  by

$$g(x) = f(x) - \frac{f(b) - f(a)}{b - a}x$$

Hence

$$\exists c \in (a, b) \text{ such that } g'(c) = 0 \quad \Leftrightarrow \quad f'(c) - \frac{f(b) - f(a)}{b - a} = 0$$

### 5.3 Proving inequalities using the mean value theorem

From the equality  $\frac{f(b) - f(a)}{b - a} = f'(c)$  in MVT,

$$m \leq f'(c) \leq M \quad \Leftrightarrow \quad m \leq \frac{f(b) - f(a)}{b - a} \leq M$$

### Example

Use MVT to show that  $e^x > 1 + x$  for all  $x > 0$

For any  $x > 0$ , define a function

$$f : [0, x] \rightarrow \mathbb{R}, \quad f(t) = e^t$$

- $f$  is continuous on  $[0, x]$
- $f$  is differentiable on  $(0, x)$

Hence, by MVT,  $\exists c \in (0, x)$  such that  $\frac{f(x) - f(0)}{x - 0} = f'(c)$

$$\Updownarrow$$

$$\frac{e^x - 1}{x} = e^c > 1$$

$$e^x - 1 > x$$

$$e^x > 1 + x$$

### Example

Use MVT to show that  $\ln(x) < x - 1$  for all  $x > 1$

For any  $x > 1$ , define a function

$$f : [1, x] \rightarrow \mathbb{R}, \quad f(t) = \ln(t)$$

By MVT,  $\exists c \in (1, x)$  such that  $\frac{f(x) - f(1)}{x - 1} = f'(c)$

$$\Updownarrow$$

$$\frac{\ln(x) - 0}{x - 1} = \frac{1}{c} < 1$$

$$\ln(x) < x - 1$$

## 5.4 Error bounds

From the equality  $\frac{f(b) - f(a)}{b - a} = f'(c)$  in MVT,

$$|f(b) - f(a)| \leq M \iff |f'(c)| |b - a| \leq M$$

### Example

How good an approximation of  $\cos 1$  is  $\cos \frac{\pi}{3}$

By MVT,  $\exists c \in (1, \frac{\pi}{3})$  such that

$$\frac{\cos(1) - \cos\left(\frac{\pi}{3}\right)}{1 - \frac{\pi}{3}} = f'(c) = -\sin(c)$$

$$\begin{aligned}\Rightarrow 0 &< \cos(1) - \cos\left(\frac{\pi}{3}\right) = -\sin(c) \left(1 - \frac{\pi}{3}\right) \\ &= \sin(c) \left(\frac{\pi}{3} - 1\right)\end{aligned}$$

$$\Rightarrow 0.5 < \cos(1) < 0.55$$

The error of approximation  $\cos(1)$  by  $\cos(\frac{\pi}{3})$  is at most 0.05

## 5.5 The sign of a derivative

Let  $f$  be a function defined on an interval  $I$ . It is said that

1.  $f$  is increasing on  $I$  if for any  $x_1, x_2 \in I$ ,

$$x_1 < x_2 \implies f(x_1) < f(x_2)$$

2.  $f$  is decreasing on  $I$  if for any  $x_1, x_2 \in I$

$$x_1 < x_2 \implies f(x_1) > f(x_2)$$

Suppose that  $f$  is continuous on  $[a, b]$  and differentiable on  $(a, b)$ .

1. If  $f'(x) > 0$  for all  $x \in (a, b)$ , then  $f$  is increasing on  $[a, b]$ .
2. If  $f'(x) < 0$  for all  $x \in (a, b)$ , then  $f$  is decreasing on  $[a, b]$ .
3. If  $f'(x) = 0$  for all  $x \in (a, b)$ , then  $f$  is constant on  $[a, b]$ .

### Proof

Want:  $f(x_1) < f(x_2)$  for all  $a \leq x_1 < x_2 \leq b$

For any  $x_1, x_2$  such that  $a \leq x_1 < x_2 \leq b$

- $f$  is continuous on  $[x_1, x_2]$
- $f$  is differentiable on  $(x_1, x_2)$

By MVT,  $\exists c \in (x_1, x_2)$  such that

$$\frac{f(x_2) - f(x_1)}{x_2 - x_1} = f'(c)$$

$$f(x_2) - f(x_1) = f'(c)(x_2 - x_1) > 0, \text{ since } f'(c), (x_2 - x_1) > 0$$

### Example

Find and classify all stationary points of the function  $f : \mathbb{R} \rightarrow \mathbb{R}$  whose derivative is given by

$$f'(x) = (x-4)(x-1)(x+5)^2$$

- Stationary points (when  $f'(x) = 0$ ):  $x = 4$ ,  $x = 1$  and  $x = -5$
- To find inflection points, check the sign of  $f'$  near the stationary point giving
  - $x = 5$ : horizontal point of inflection
  - $x = 1$ : local maximum point
  - $x = 4$ : local minimum point

## 5.6 The second derivative and applications

Suppose that a function  $f$  is twice differentiable on  $(a, b)$  and that  $c \in (a, b)$ .

1. If  $f'(c) = 0$  and  $f''(c) > 0$  then  $c$  is a local minimum point of  $f$ .
2. If  $f'(c) = 0$  and  $f''(c) < 0$  then  $c$  is a local maximum point of  $f$ .

### Facts

Suppose that  $g$  is differentiable at a point  $c$

1. If  $g'(c) > 0$ , then  $g(c-h) < g(c) < g(c+h)$  for all sufficiently small  $h > 0$ .
2. If  $g'(c) < 0$ , then  $g(c+h) < g(c) < g(c-h)$  for all sufficiently small  $h > 0$ .

Proof of 1.

$$g'(c) = \lim_{\delta \rightarrow 0} \frac{1}{\delta} (g(c+\delta) - g(c))$$
$$g(c+\delta) - g(c) \approx g'(c)\delta, \quad \text{for sufficiently small } \delta$$

- $\delta > 0 : g'(c)\delta > 0 \Rightarrow g(c+|\delta|) - g(c) > 0$  if  $\delta = h$  is sufficiently small
- $\delta < 0 : g'(c)\delta < 0 \Rightarrow g(c+\delta) = g(c+|\delta|) - g(c) > 0$  if  $|\delta| = h$  is sufficiently small

If  $f'(c)$  and  $f''(c) = 0$ , then the second derivative test cannot be applied. Here, it is best to check the sign of the derivative on either side of  $c$ .

## 5.7 Critical points, maxima and minima

Let  $f$  be defined on  $[a, b]$ . Then  $c \in [a, b]$  is a critical point for  $f$  on  $[a, b]$  if

1.  $c$  is an endpoint ( $c = a$  or  $c = b$ ), or
2.  $f$  is not differentiable at  $c$ , or
3.  $f$  is differentiable at  $c$  and  $f'(c) = 0$

Suppose that  $f$  is continuous on  $[a, b]$ . Then

1.  $f$  has a global maximum and a global minimum on  $[a, b]$ , and
2. the global maximum point and the global minimum point are both critical points for  $f$  on  $[a, b]$ .

### Example

Suppose that  $f : \mathbb{R} \rightarrow \mathbb{R}$  is given by the rule

$$f(x) = |x^2 - 3x - 4|.$$

Find the global maximum and global minimum values of  $f$  on the interval  $[0, 5]$

$$f(x) = \begin{cases} -(x^2 - 3x - 4) = -(x - 4)(x + 1), & x \in [0, 4) \\ x^2 - 3x - 4 = (x - 4)(x + 1), & x \in [4, 5] \end{cases}$$

- Endpoints:
  - $f(0) = 4$
  - $f(5) = 6$
- Non-differentiable points:  $f(4) = 0$
- Stationary points:

$$\begin{aligned} \frac{d}{dx}(x^2 - 3x - 4) &= 2x - 3 = 0 \\ x &= \frac{3}{2} \\ f\left(\frac{3}{2}\right) &= \frac{25}{4} = 6.25 \end{aligned}$$

- Global maximum at  $x = \frac{3}{2}$ . Global minimum at  $x = 4$

## 5.8 Counting zeros

### Example

Determine the number of real zeroes of  $f : \mathbb{R} \rightarrow \mathbb{R}$ ,

$$f(x) = x^4 - x^3 - 3x^2 - 8x - 5$$

and give an approximate location for each zero.

1. Differentiate

$$f'(x) = 4x^3 - 3x^2 - 6x - 8 = (x - 2)(4x^2 + 5x + 4)$$

2. Stationary point and signs:

$$\begin{aligned} f'(2) &= 0, & f'(x) &< 0 \text{ on } (-\infty, 2) \\ & & f'(x) &> 0 \text{ on } (2, \infty) \end{aligned}$$

3. Apply IVT:

- At  $x = 2, f(2) = -25$
- On  $(-\infty, 2)$ ,  $f$  is decreasing  $\Rightarrow$  at most one zero on  $(-\infty, 2)$   
 $f(-1) = 2, f(0) = -5 \Rightarrow$  at least one zero on  $(-1, 0)$  by IVT.
- On  $(2, \infty)$ ,  $f$  is increasing  $\Rightarrow$  at most one zero on  $(2, \infty)$   
 $f(3) = -2, f(4) = 107 \Rightarrow$  at least one zero on  $(3, 4)$  by IVT.

4. Hence,  $f$  has 2 real zeros



## 5.9 Antiderivatives

Suppose that  $f$  is continuous on an open interval  $I$ . A function  $F$  is an antiderivative (or a primitive) of  $f$  on  $I$  if

$$F'(x) = f(x), \quad \forall x \in I.$$

Let  $f$  be a continuous function on an open interval  $I$ . Suppose that  $F$  and  $G$  are two antiderivatives of  $f$  on  $I$ . Then there is  $C \in \mathbb{R}$  such that

$$G(x) = F(x) + C, \quad \forall x \in I.$$

### Proof for non-uniqueness of antiderivatives

Let  $H : I \rightarrow \mathbb{R}$  be given by  $H(x) = G(x) - F(x)$ .

Then  $H$  is differentiable on  $I$ , with

$$\begin{aligned} H'(x) &= G'(x) - F'(x) \\ &= f(x) - f(x) \\ &= 0, \quad \forall x \in I. \end{aligned}$$

So,  $H(x)$  is a constant for all  $x \in I$ . Hence,  $H(x) = G(x) - F(x) = C, \quad \forall x \in I$

## 5.10 L'Hopital's rule

Let  $f$  and  $g$  be both differentiable in a neighbourhood of some  $a \in \mathbb{R}$ . If

1.  $f(x) \rightarrow 0$  and  $g(x) \rightarrow 0$  as  $x \rightarrow a$ , or
2.  $f(x) \rightarrow \infty$  and  $g(x) \rightarrow \infty$  as  $x \rightarrow a$ , or

then

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}, \quad \text{provided the limit } \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)} \text{ exists.}$$

The same rule applies to one-sided limits and the cases  $x \rightarrow \infty$  and  $x \rightarrow -\infty$

### Proof

Since  $f$  and  $g$  are differentiable at  $a$ , they must be continuous at  $a$ .

$$\begin{aligned} &\implies \lim_{x \rightarrow a} f(x) = 0 = f(a) \quad \text{and} \quad \lim_{x \rightarrow a} g(x) = 0 = g(a) \\ &\implies \frac{f(x)}{g(x)} = \frac{f(x) - f(a)}{g(x) - g(a)} = \frac{f'(c)(x-a)}{g'(d)(x-a)} \quad \text{for some } c \text{ and } d \text{ between } a \text{ and } x. \\ &\implies \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(c)}{g'(d)} = \frac{f'(a)}{g'(a)} \end{aligned}$$

### Example

1. Determine  $\lim_{x \rightarrow 0} \frac{x \sin(x)}{1 - \cos(x)}$ , (indeterminate form  $\frac{0}{0}$ )

$$\begin{aligned}\lim_{x \rightarrow 0} \frac{x \sin(x)}{1 - \cos(x)} &= \lim_{x \rightarrow 0} \frac{\sin(x) + x \cos(x)}{\sin(x)} \\ &= \lim_{x \rightarrow 0} 1 + \frac{x}{\sin(x)} \cos(x) \\ &= 1 + \lim_{x \rightarrow 0} \frac{x}{\sin(x)} \lim_{x \rightarrow 0} \cos(x) = 1 + 1 \times 1 = 2\end{aligned}$$

2. Determine  $\lim_{x \rightarrow 0} \frac{\sin(x) - x \cos(x)}{x - \sin(x)}$ , (indeterminate form  $\frac{0}{0}$ )

$$\begin{aligned}\lim_{x \rightarrow 0} \frac{\sin(x) - x \cos(x)}{x - \sin(x)} &= \lim_{x \rightarrow 0} \frac{\cos(x) + x \sin(x) - \cos(x)}{1 - \cos(x)} \\ &= \lim_{x \rightarrow 0} \frac{x \sin(x)}{1 - \cos(x)} \\ &= 2, \quad \text{by part 1}\end{aligned}$$

## Chapter 6

# Inverse functions

### 6.1 Some preliminary examples

Given  $f : A \rightarrow B$ ,  $x \mapsto f(x) = y$ . Can we express  $x$  in terms of  $y$ ?  $g : C \rightarrow A, y \mapsto g(y) = x$

#### Example

Can be invert  $f : [0, \infty) \rightarrow \mathbb{R}$  given by  $f(x) = \frac{x}{x+1}$

$$\begin{aligned}\text{Start from } y &= \frac{x}{x+1} \\ \Rightarrow y(x+1) &= x \\ \Rightarrow xy + y &= x \\ \Rightarrow x(y-1) &= -y \\ \Rightarrow x &= -\frac{y}{y-1}, \quad \text{where } y \in [0, 1)\end{aligned}$$

Since for every  $y \in [0, 1)$ , there is only one  $x$ ,  $f$  can be inverted

### 6.2 One-to-one functions

A function  $f$  is said to be one-to-one (or injective) if

$$f(x_1) = f(x_2) \implies x_1 = x_2, \quad \forall x_1, x_2 \in \text{Dom}(f),$$

or equivalently

$$f(x_1) \neq f(x_2) \implies x_1 \neq x_2, \quad \forall x_1, x_2 \in \text{Dom}(f),$$

### Example

For which  $\alpha \in \mathbb{R}$  is the function  $f : \mathbb{R} \ni x \mapsto x^3 + \alpha x + 1 \in \mathbb{R}$  one-to-one?

Assume  $f(x_1) = f(x_2)$ . Show that  $x_1 = x_2$

$$\begin{aligned}x_1^3 + \alpha x_1 + 1 &= x_2^3 + \alpha x_2 + 1, \quad x_1, x_2 \in \mathbb{R} \\ \Rightarrow x_1^3 - x_2^3 + \alpha(x_1 - x_2) &= 0 \\ \Rightarrow (x_1 - x_2)(x_1^2 + x_1x_2 + x_2^2 + \alpha) &= 0 \\ \frac{3}{4}(x_1 + x_2)^2 + \frac{1}{4}(x_1 - x_2)^2 &\end{aligned}$$

- Case 1:  $\alpha = 0$   
 $\Rightarrow (x_1 - x_2) \left[ \frac{3}{4}(x_1 + x_2)^2 + \frac{1}{4}(x_1 - x_2)^2 \right] = 0$   
 $\Rightarrow x_1 = x_2$   
 $\Rightarrow f$  is one-to-one
- Case 2:  $\alpha > 0$

...

### Conditions for one-to-one

Horizontal line test

Let  $A, B \subseteq \mathbb{R}$ . The graph of a function  $f : A \rightarrow B$  is

$$\{(x, f(x)) | x \in A\}$$

$f$  is one-to-one  $\iff$  every horizontal line cuts the graph of  $f$  at most once

- Not every one-to-one function is strictly monotone, (Continuous one-to-one  $f : I \rightarrow \mathbb{R}$  is strictly monotone)

## 6.3 Inverse functions

Suppose that  $f$  is a one-to-one function. Then there exists a unique function  $g$  such that

- $(g \circ f)(x) = g(f(x)) = x$ , for all  $x \in \text{Dom}(f)$
- $(f \circ g)(x) = f(g(x)) = x$ , for all  $y \in \text{Range}(f)$
- $\text{Dom}(g) = \text{Range}(f)$
- $\text{Range}(g) = \text{Dom}(f)$
- $g$  is one-to-one

Suppose that  $f$  is a one-to-one function. The inverse function of  $f$  is the unique function  $g$  given by the above theorem, often denoted by

$$f^{-1} : \text{Range}(f) \rightarrow \text{Dom}(f).$$

Note:

- $f^{-1}$  is not  $\frac{1}{f}$
- $f^{-1}$  is a function that can be graphed

## 6.4 The inverse function theorem

Let  $f : \mathbb{R} \rightarrow \mathbb{R}$  be given by  $f(x) = x^3$ .

- $f$  is differentiable everywhere
- its inverse  $f^{-1} : \mathbb{R} \rightarrow \mathbb{R}$  is given by  $f^{-1}(x) = \sqrt[3]{x}$

$$(f^{-1})'(x) = \frac{1}{3\sqrt[3]{x^2}} \implies \text{not differentiable at } x = 0$$

Let  $I$  be an open interval. Suppose that  $f : I \rightarrow \mathbb{R}$  is differentiable and  $f'(x) \neq 0$  for all  $x \in I$ . Then

1.  $f$  is one-to-one and has an inverse  $g : \text{Range}(f) \rightarrow \text{Dom}(f)$
2.  $g$  is differentiable at all points in  $\text{Range}(f)$

$$g'(y) = \frac{1}{f'(g(y))}, \quad \forall y \in \text{Range}(f).$$

The derivative of an inverse function can hence be determined without finding the inverse function.

### Example

- For the function  $f : (0, \infty) \rightarrow (0, \infty)$  given by  $f(x) = x^3$ , its inverse is

$$g : (0, \infty) \rightarrow (0, \infty), \quad g(x) = \sqrt[3]{x}.$$

Then

$$g'(x) = \frac{1}{f'(g(x))} = \frac{1}{3[g(x)]^2} = \frac{1}{3x^{\frac{2}{3}}}$$

- Determine the derivative of the inverse of the function

$$f : (-1, 1) \ni x \mapsto 3 + x^2 + 2 \tan\left(\frac{\pi}{2}x\right) \in \mathbb{R} \quad \text{at } x = 3.$$

Using IFT,

$$\begin{aligned} g'(x) &= \frac{1}{f'(g(x))} \\ \Rightarrow g'(3) &= \frac{1}{f'(g(3))} \quad \text{where } 3 = f(0) \Rightarrow g(3) = 0 \\ \Rightarrow g'(3) &= \frac{1}{f'(0)} = \frac{1}{2 \cdot 0 + \pi \sec^2(\frac{\pi}{2} \cdot 0)} = \frac{1}{\pi} \end{aligned}$$

## 6.5 Applications to the trigonometric functions

- Since  $\sin : \mathbb{R} \rightarrow [-1, 1]$  is not one to one, we restrict its domain. The restricted sine function  $\sin : [-\frac{\pi}{2}, \frac{\pi}{2}] \rightarrow [-1, 1]$  is one-to-one, and has an inverse  $\sin^{-1} : [-1, 1] \rightarrow [-\frac{\pi}{2}, \frac{\pi}{2}]$ . Any restriction on sine can be used as long as it is a one-to-one function.
- On  $(-1, 1)$ , by Inverse Function Theorem,

$$\begin{aligned} \frac{d}{dx} (\sin^{-1} x) &= \frac{1}{\cos(\sin^{-1} x)} \\ &= \frac{1}{\sqrt{1 - (\sin(\sin^{-1} x))^2}} = \frac{1}{\sqrt{1 - x^2}} \quad (\rightarrow \infty \text{ as } x \rightarrow \pm 1) \end{aligned}$$

Example

- Determine  $\cos(2\sin^{-1}\frac{3}{5})$

# Chapter 7

## Curve sketching

### 7.1 Curves defined by a Cartesian equation

#### 7.1.1 A checklist for sketching curves

- Identify the domain of  $f$
- Identify any symmetries:
  - $f$  is odd if  $f(-x) = -f(x)$  for all  $\pm x \in \text{Dom}(f)$
  - $f$  is even if  $f(-x) = f(x)$  for all  $\pm x \in \text{Dom}(f)$
  - $f$  is periodic if  $f(x + T) = f(x)$  for all  $\pm x, x + T \in \text{Dom}(f)$  (period =  $T$ )
- Find x- and y-axis intercepts
- Identify vertical asymptotes
- Examine the behaviour of  $f(x)$  as  $x \rightarrow \pm\infty$
- Use calculus to identify stationary points and other features.

#### 7.1.2 Oblique asymptotes

Suppose that  $a, b \in \mathbb{R}$  and  $a \neq 0$ . We say that a straight line  $y = ax + b$  is an oblique asymptote for a function  $f$  if

$$\lim_{x \rightarrow \pm\infty} (f(x) - (ax + b)) = 0 \quad \text{or} \quad \lim_{x \rightarrow \mp\infty} (f(x) - (ax + b)) = 0.$$

### Example

Find the oblique asymptotes to the function  $f$  defined by

$$f(x) = \frac{(x-1)|x| + x}{x-2}, \quad \forall x \neq 2.$$

- If  $x > 0$ , then

$$\begin{aligned} f(x) &= \frac{(x-2)x + x}{x-2} = x + \frac{x}{x-2} \\ &= x + 1 + \frac{2}{x-2} \end{aligned}$$

the line  $y = x + 1$  is an oblique asymptote

- If  $x \leq 0$ , then

$$\begin{aligned} f(x) &= \frac{-(x-2)x + x}{x-2} = -x + \frac{x}{x-2} \\ &= -x + 1 + \frac{2}{x-2} \end{aligned}$$

the line  $y = x - 1$  is an oblique asymptote

## 7.2 Parametrically defined curves

In a plane, parametrically defined curves are in the form

$$(x(t), y(t)), \quad t \in A$$

- $t$  is the parameter
- $A$  is a given domain

### 7.2.1 Calculus and parametric curves

Given a curve  $\gamma(t) = (x(t), y(t))$

- $\gamma(t)$  is a position of a particle at time  $t$
- $\gamma'(t) = (x'(t), y'(t))$  is velocity of the particle at time  $t$
- gradient or slope of the curve at  $t$ : derive from  $\frac{\text{rise}}{\text{run}}$

### 7.2.2 The cycloid and curve of fastest descent

The curve of fastest descent between two points is defined by a unique cycloid.

## 7.3 Curves defined by polar coordinates

### 7.3.1 Polar coordinates

Every point  $P$  in the plane can be specified by the distance  $r$  of  $P$  from  $O$  and  $\theta$ , the angle anticlockwise between  $OP$  and the positive horizontal axis. Polar coordinates  $(r, \theta)$  are related to the Cartesian coordinates  $(x, y)$  by

$$\begin{aligned} x &= r \cos \theta, \quad y = r \sin \theta, \\ r &= \sqrt{x^2 + y^2}, \quad \tan \theta = \frac{y}{x} \text{ (provided } x \neq 0) \end{aligned}$$



### 7.3.2 Basic sketches of polar curves

Many curves can be described by equations of the form

$$r = f(\theta) \quad \text{or} \quad \theta = g(r)$$

$\Rightarrow$  parametrically defined curves:

$$\gamma(\theta) = (f(\theta)\cos\theta, f(\theta)\sin\theta) \quad \text{or} \quad \gamma(r) = (r\cos g(r), r\sin g(r))$$

#### Symmetries

Curves described by  $r = f(\theta)$

- $f$  is an even function  $\Rightarrow$  polar is symmetric about  $x$ -axis
- $f(\pi - \theta) = f(\theta) \Rightarrow$  polar curve is symmetric about  $y$ -axis

#### Example

Sketch the curve described by the polar equation  $r = 2\sin\theta$ ,  $0 \leq \theta \leq \pi$ .

$$\begin{aligned} r = 2\sin\theta &\Rightarrow r^2 = 2r\sin\theta \\ &\Rightarrow x^2 + y^2 = 2y \\ &\Rightarrow x^2 + (y-1)^2 - 1 = 0 \end{aligned}$$

### 7.3.3 Sketching polar curves using calculus

Given a curve in the polar form  $r = f(\theta)$ .

- Its parametric form:

$$\begin{aligned} \gamma(\theta) &= (x(\theta), y(\theta)) \\ &= (f(\theta)\cos\theta, f(\theta)\sin\theta) \end{aligned}$$

- Tangent vector:

$$\begin{aligned} \gamma'(\theta) &= (x'(\theta), y'(\theta)) \\ &= (f'(\theta)\cos\theta - f(\theta)\sin\theta, f'(\theta)\sin\theta + f(\theta)\cos\theta) \end{aligned}$$

**Note:** It is often convenient to allow negative values of  $r$ .

$$\begin{aligned} x &= r\cos\theta = (-r)\cos(\theta + \pi), \\ y &= r\sin\theta = (-r)\sin(\theta + \pi) \end{aligned}$$

# Chapter 8

## Integration

### 8.1 Area and the Riemann Integral

#### 8.1.1 Area of regions with curved boundaries

Any definition of area should satisfy:

1. If  $\Omega$  is a region of the plane, then

$$\text{area}(\Omega) \geq 0.$$

2. If one region  $\Omega_1$  is contained in another region  $\Omega_2$ , then

$$\text{area}(\Omega_1) \leq \text{area}(\Omega_2).$$

3. If the area of a region  $\Omega$  is partitioned into two smaller disjoint regions  $\Omega_1$  and  $\Omega_2$ , then

$$\text{area}(\Omega) = \text{area}(\Omega_1) + \text{area}(\Omega_2).$$

4. If  $\Omega_1$  and  $\Omega_2$  are congruent regions, then

$$\text{area}(\Omega_1) = \text{area}(\Omega_2).$$

5. If  $\Omega$  is a rectangle of length  $a$  and height  $b$ , then

$$\text{area}(\Omega) = ab.$$

#### 8.1.2 Approximations of area using Riemann sums

Consider the function  $f : \mathbb{R} \rightarrow \mathbb{R}$  given by  $f(x) = x^2$ .

Idea: approximate  $\text{area}(\Omega)$  with sums of areas of  $n$  rectangles.

- $\text{area}(\Omega_1) \leq \text{area}(\Omega) \leq \text{area}(\Omega_2)$

To find  $\text{area}(\Omega)$ :

1. Divide  $[0, 1]$  into  $n$  subintervals  $\left[0, \frac{1}{n}\right], \left[\frac{1}{n}, \frac{2}{n}\right], \left[\frac{2}{n}, \frac{3}{n}\right], \dots, \left[\frac{n-1}{n}, 1\right]$

The set  $\mathcal{P}_n = \left\{0, \frac{1}{n}, \frac{2}{n}, \frac{3}{n}, \dots, \frac{n-1}{n}, 1\right\}$

2. Find bounds for  $\text{area}(\Omega)$ . An upper bound:

$$\begin{aligned}\bar{R}_k &= \frac{1}{n} \times f\left(\frac{k}{n}\right) = \frac{k^2}{n^3} \\ \sum_{k=1}^n \bar{R}_k &= \sum_{k=1}^n \frac{k^2}{n^3} \\ &= \frac{1}{n^3} \times \frac{1}{6} n(n+1)(2n+1) \\ &= \frac{1}{3} + \frac{1}{2n} + \frac{1}{6n^2}\end{aligned}$$

$\text{area}(\Omega) \leq \bar{S}_{\mathcal{P}_n}(f) := \sum_{k=1}^n \bar{R}_k$  the upper Riemann sum of  $f$  with respect to  $\mathcal{P}_n$

A lower bound

$$\begin{aligned}\underline{R}_k &= \frac{1}{n} \times f\left(\frac{k-1}{n}\right) = \frac{(k-1)^2}{n^3} \\ \sum_{k=1}^n \underline{R}_k &= \sum_{k=1}^n \frac{(k-1)^2}{n^3} \\ &= \frac{1}{3} - \frac{1}{2n} + \frac{1}{6n^2}\end{aligned}$$

$\text{area}(\Omega) \geq \underline{S}_{\mathcal{P}_n}(f) := \sum_{k=1}^n \underline{R}_k$  the lower Riemann sum of  $f$  with respect to  $\mathcal{P}_n$

3. Take the limit:

$$\lim_{n \rightarrow \infty} \bar{S}^{\mathcal{P}_n} = \lim_{n \rightarrow \infty} \underline{S}^{\mathcal{P}_n} = \frac{1}{3} \implies A = \frac{1}{3}$$

The process of calculating upper and lower Riemann sums and taking a limit is called integration.

### 8.1.3 Definitions: area under the graph of a function and the Riemann integral

A finite set  $\mathcal{P}$  of points in  $\mathbb{R}$  is said to be a partition of  $[a, b]$  if

1.  $\mathcal{P} = \{a_0, a_1, a_2, \dots, a_n\}$ , and
2.  $a = a_0 < a_1 < a_2 < \dots < a_n = b$ .

Note that the points of  $\mathcal{P}$  need not be evenly spaced.

#### Area under graph of $f$

Suppose that  $f$  is a bounded function on  $[a, b]$ , and  $f(x) \geq 0$  for all  $x \in [a, b]$ .

- An upper bound:
  - $\bar{f}_k$  = Maximum value of  $f$  on the subinterval  $[a_{k-1}, a_k]$
  - area of the  $k$ th rectangle =  $(a_k - a_{k-1})\bar{f}_k$
- A lower bound:
  - $\underline{f}_k$  = Minimal value of  $f$  on the subinterval  $[a_{k-1}, a_k]$
  - area of the  $k$ th rectangle =  $(a_k - a_{k-1})\underline{f}_k$

Suppose that a function  $f$  is bounded on  $[a, b]$  and  $f(x) \geq 0$  for all  $x \in [a, b]$ . If there exists a unique  $A \in \mathbb{R}$  such that

$$\underline{S}_{\mathcal{P}}(f) \leq A \leq \overline{S}_{\mathcal{P}}(f)$$

Suppose that a function  $f$  is bounded on  $[a, b]$ . If there exists a unique  $I \in \mathbb{R}$  such that

$$\underline{S}_{\mathcal{P}}(f) \leq I \leq \overline{S}_{\mathcal{P}}(f) \quad \text{for every partition } \mathcal{P} \text{ of } [a, b],$$

then

- we say that  $f$  is Riemann integrable on the interval  $[a, b]$ ,
- we call  $I$  the definite integral of  $f$  from  $a$  to  $b$ , and write  $I = \int_a^b f(x)dx$ .

## 8.2 Integration using Riemann sums

### Example

Suppose that  $f(x) = C > 0$ . Show that  $f$  is Riemann integrable on  $[a, b]$  and calculate the area under the graph of  $f$  from  $a$  to  $b$ .

For any partition  $\mathcal{P} = \{a_0, a_1, \dots, a_n\}$  of  $[a, b]$ ,

$$\begin{aligned} \overline{S}_{\mathcal{P}}(f) &= \sum_{k=1}^n \bar{f}_k (a_k - a_{k-1}) = C \sum_{k=1}^n (a_k - a_{k-1}) = C(b - a) \\ \underline{S}_{\mathcal{P}}(f) &= \sum_{k=1}^n f_k (a_k - a_{k-1}) = \overline{S}_{\mathcal{P}}(f) \end{aligned}$$

$\Rightarrow f$  is Riemann integrable

$$\Rightarrow \int_a^b f(x)dx = C(b - a)$$

A function  $f : [a, b] \rightarrow \mathbb{R}$  is said to be piecewise continuous if it is continuous on  $[a, b]$  at all except perhaps a finite number of points.

If a function  $f$  is bounded and piecewise continuous on  $[a, b]$ , then  $f$  is Riemann integrable on  $[a, b]$ .

Note: Given that  $f$  is Riemann integrable on  $[a, b]$ , to find its definite integral, it is sufficient to find some partition  $\mathcal{P}_n$  of  $[a, b]$  such that

$$\lim_{n \rightarrow \infty} \overline{S}_{\mathcal{P}_n}(f) = \lim_{n \rightarrow \infty} \underline{S}_{\mathcal{P}_n}(f)$$

Then  $\int_a^b f(x)dx = I = \text{the common limit.}$

### Example

Find  $\int_1^4 x^4 dx$  using a "clever" partition.

$$\alpha = 2^{\frac{1}{n}} > 1.$$

$$\alpha_0 = 1$$

$$\alpha_n = (2^{\frac{1}{n}})^n = 2$$

$$\begin{aligned} \underline{S}_P &= (\alpha - 1) \cdot 1^4 + (\alpha^2 - \alpha)\alpha^4 + (\alpha^3 - \alpha^2)(\alpha^2)^4 + \cdots + (\alpha^n - \alpha^{n-1})(\alpha^{n-1})^4 \\ &= (\alpha - 1) [1 + \alpha^5 + \alpha^{10} + \cdots + (\alpha^{n-1})^5] \\ &= (\alpha - 1) \sum_{k=0}^{n-1} \alpha^{5k} \\ &= (\alpha - 1) \frac{1 \cdot (1 - \alpha^{5n})}{1 - \alpha^5} \quad (S_n = \frac{1}{1-r} a(r - r^{n+1})) \\ &= \frac{\alpha^{5n} - 1}{\alpha^4 + \alpha^3 + \alpha^2 + \alpha + 1} \\ &= \frac{2^5 - 1}{\alpha^4 + \alpha^3 + \alpha^2 + \alpha + 1} \quad (\text{since } \alpha^{5n} = (2^{\frac{1}{n}})^{5n} = 2^5) \\ &\rightarrow \frac{31}{5} \text{ as } n \rightarrow \infty \quad (\text{since } \alpha \rightarrow 1 \text{ as } n \rightarrow \infty) \\ \overline{S}_P &= (\alpha - 1)\alpha^4 + (\alpha^2 - \alpha)(\alpha^2)^4 + (\alpha^3 - \alpha^2)(\alpha^3)^4 + \cdots + (\alpha^n - \alpha^{n-1})(\alpha^n)^4 \\ &= \alpha^4 \underline{S}_P \\ &\rightarrow \frac{31}{5} \text{ as } n \rightarrow \infty \\ \Rightarrow I &= \frac{31}{5} \end{aligned}$$

## 8.3 The Riemann integral and signed area

$$\begin{aligned} \int_a^c f(x) dx &= \text{area}(\Omega_1) - \text{area}(\Omega_2) \\ &= \text{Signed area under the graph of } f \text{ from } a \text{ to } c \quad \int_a^c |f(x)| dx = \text{area}(\Omega_1) + \text{area}(\Omega_2) \\ &= \text{Unsigned area under the graph of } f \text{ from } a \text{ to } c \end{aligned}$$

## 8.4 Basic properties of the Riemann integral

Suppose that  $f, g : a, b \rightarrow \mathbb{R}$  are integrable.

- $\alpha f + \beta g$  is integrable for any  $\alpha, \beta \in \mathbb{R}$ , and

$$\int_a^b (\alpha f + \beta g)(x) dx = \alpha \int_a^b f(x) dx + \beta \int_a^b g(x) dx.$$

- If  $a < c < b$ , then

$$\int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx.$$

- If  $f(x) \geq 0$  for all  $x \in [a, b]$ , then  $\int_a^b f(x)dx \geq 0$ .
- If  $f(x) \leq g(x)$  for all  $x \in [a, b]$ , then

$$\int_a^b f(x)dx \leq \int_a^b g(x)dx.$$

- If  $m \leq f(x) \leq M$  for all  $x \in [a, b]$ , then

$$m(b-a) \leq \int_a^b f(x)dx \leq M(b-a)$$

- $|f|$  is integrable on  $[a, b]$ , and

$$\left| \int_a^b f(x)dx \right| \leq \int_a^b |f(x)|dx.$$

Suppose that  $b < a$  and that  $f$  is integrable on  $[b, a]$ . Then we define

$$\int_a^b f(x)dx := - \int_b^a f(x)dx,$$

and

$$\int_a^a f(x)dx := 0.$$

Suppose that  $a, b, c \in \mathbb{R}$  and that  $f$  is integrable over some interval containing  $a, b$  and  $c$ . Then

$$\int_a^b f(x)dx = \int_a^c f(x)dx + \int_c^b f(x)dx.$$

## 8.5 The first fundamental theorem of calculus

For a continuous function  $f$  on  $[a, b]$ , it is integrable on  $[a, b]$ , and the function  $F : [a, b] \rightarrow \mathbb{R}$  given by

$$F(x) = \int_a^x f(t)dt, \quad \forall x \in [a, b]$$

is the signed area on the interval  $[a, x]$

- If  $f(t) > 0$  for all  $t \in [a, b]$ , then  $F$  is an increasing function
- Differentiability:

$$\frac{F(x+h) - F(x)}{h} \rightarrow ?$$

If  $f : [a, b] \rightarrow \mathbb{R}$  is a continuous function, then the function  $F : [a, b] \rightarrow \mathbb{R}$  defined by  $F(x) = \int_a^x f(t)dt$  is

- continuous on  $[a, b]$ , and
- differentiable on  $(a, b)$  with derivative  $F'$  given by

$$F'(x) = f(x), \quad \forall x \in (a, b)$$

Note:

- Every continuous  $f$  has an antiderivative  $F$  given by  $F(x) = \int_a^x f(t)dt$
- Every antiderivative of  $f$  is of the form  $F + \text{constant}$
- Integrate, then differentiate gives back  $f$
- Differentiate then integrate may not give back  $f$

## 8.6 The second fundamental theorem of calculus

Suppose that  $f$  is a continuous function on  $[a, b]$ . Let  $F$  be a continuous function on  $(a, b)$  such that  $F$  is an antiderivative of  $f$  on  $(a, b)$ . Then

$$\int_a^b f(t)dt = F(b) - F(a)$$

### 8.6.1 Variable limits of integration

Suppose that  $f$  is continuous and  $a, b$  are differentiable. Let  $I(x) = \int_{a(x)}^{b(x)} f(t)dt$ . Let  $F$  be an antiderivative of  $f$ . Then

$$\begin{aligned} I(x) &= F(b(x)) - F(a(x)) \\ \frac{dI(x)}{dx} &= F'(b(x))b'(x) - F'(a(x))a'(x) \\ &= f(b(x))b'(x) - f(a(x))a'(x). \end{aligned}$$

#### Example

$$\text{Let } I(x) = \int_x^{x^2} f(t)dt, f(t) = \begin{cases} \frac{\sin t}{t}, & t \neq 0 \\ 1, & t = 0 \end{cases}. \text{ Find } I'(x)$$

$$\begin{aligned} I'(x) &= f(b(x))b'(x) - f(a(x))a'(x) \quad \text{where } a(x) = x, b(x) = x^2 \\ &= 2x f(x^2) - f(x) \\ &= \begin{cases} 2 \frac{\sin(x^2)}{x} - \frac{\sin(x)}{x} & \text{for } x \neq 0 \\ 2 \cdot 0 \cdot f(0) - f(0) = -1 & \text{for } x = 0. \end{cases} \end{aligned}$$

## 8.7 Indefinite integrals

An indefinite integral is an integral without limits  $\int f(t)dt$ .

$$\int f(t)dt = F(t) + C, \quad C \text{ is a constant of integration.}$$

## 8.8 Integration by substitution

Identify the integrand in the form  $f(g(x))g'(x)$  to make the substitution  $u = g(x)$ , so that  $du = g'(x)dx$

### 8.8.1 Cases with symmetry

Suppose that  $f$  is a continuous function and  $a \in \mathbb{R}$ .

- If  $f$  is even, then  $\int_{-a}^a f(x)dx = 2 \int_0^a f(x)dx$ .
- If  $f$  is odd, then  $\int_{-a}^a f(x)dx = 0$

## 8.9 Integration by parts

Example

$$\int x \cos x dx = \int x \left( \frac{d}{dx} \sin(x) \right) dx = x \sin x - \int 1 \cdot \sin(x) dx = x \sin x + \cos x + c$$

## 8.10 Improper integrals

Integrate over  $[a, \infty)$  instead of  $[a, b]$

- Suppose that there is  $L \in \mathbb{R}$  such that  $\int_a^R f(x) dx \rightarrow L$  as  $R \rightarrow \infty$ . Then we say
  - $f$  is integrable over  $[a, \infty)$
  - the improper integral  $\int_a^\infty f(x) dx$  is convergent, and write  $\int_a^\infty f(x) dx = L$ .
- Suppose that  $\int_a^R f(x) dx$  does not have a limit as  $R \rightarrow \infty$ . Then we say
  - $f$  is not integrable over  $[a, \infty)$ ,
  - the integral  $\int_a^\infty f(x) dx$  is divergent
- If  $f$  is integrable over both  $(-\infty, 0]$  and  $[0, \infty)$ , then we say that  $f$  is integrable over  $(-\infty, \infty)$  and write

$$\int_{-\infty}^\infty f(x) dx = \int_{-\infty}^0 f(x) dx + \int_0^\infty f(x) dx.$$

- If  $f$  is not integrable on either of the intervals  $(-\infty, 0]$  or  $[0, \infty)$ , then we say that the improper integral  $\int_{-\infty}^\infty f(x) dx$  diverges.

## 8.11 Comparison tests for improper integrals

Let  $f$  and  $g$  be integrable on finite intervals and  $0 \leq f(x) \leq g(x)$  for all  $x > a$ .

- If  $\int_a^\infty g(x) dx$  converges, then  $\int_a^\infty f(x) dx$  converges
- If  $\int_a^\infty f(x) dx$  diverges, then  $\int_a^\infty g(x) dx$  diverges

Let  $f$  and  $g$  be non-negative and bounded on  $[a, \infty)$ . If  $\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = L \in (0, \infty)$ , then  $\int_a^\infty f(x) dx$  and  $\int_a^\infty g(x) dx$  either both converge or diverge.

For  $f : [a, b] \rightarrow \mathbb{R}$ , we may also define

$$\int_a^b f(x) dx := \lim_{R \rightarrow b^-} \int_a^R f(x) dx, \quad (\text{provided the limit exists})$$



## Chapter 9

# The logarithmic and exponential functions

### 9.1 Powers and logarithms

**Definition.** Suppose that  $b > 0$  and  $b \neq 1$ ,  $c \in \mathbb{Q}$  and  $a = b^c$ . Then

$$c = \log_b a.$$

**Properties.** If  $x > 0$ ,  $y > 0$ ,  $b > 0$  with  $b \neq 1$  and  $r \in \mathbb{Q}$ , then

- $\log_b 1 = 0$ ,  $\log_b b = 1$
- $\log_b(xy) = \log_b x + \log_b y$ ,  $\log_b \frac{x}{y} = \log_b x - \log_b y$
- $\log_b(b^r) = r$ ,  $\log_b(x^r) = r \log_b x$ .

### 9.2 The natural logarithm function

**Definition.** The function  $\ln : (0, \infty) \rightarrow \mathbb{R}$  is defined by the formula

$$\ln(x) = \int_1^x \frac{1}{t} dt$$

### 9.3 The exponential function

**Definition:** The function  $\exp : \mathbb{R} \rightarrow (0, \infty)$  is defined to be the inverse function of  $\ln : (0, \infty) \rightarrow \mathbb{R}$ . For a real  $x \notin \mathbb{Q}$ , we define the real number  $e^x$  by

$$e^x = \exp x.$$

### 9.4 Exponentials and logarithms with other bases

For any  $r \in \mathbb{Q}$  and real number  $b > 0$ ,

$$b^r = \exp(\ln(b^r)) = \exp(r \ln b) = e^{r \ln b}.$$

**Definition:** For any real  $x \notin \mathbb{Q}$  and  $b > 0$ , we define the number  $b^x$  by

$$b^x = \exp(x \ln b) = e^{x \ln b}.$$

## 9.5 Integration and the $\ln$ function

Suppose that  $x \neq 0$ .

$$\frac{d}{dx}(\ln(x)) = \begin{cases} \frac{1}{x} & \text{if } x > 0 \\ \frac{1}{-x} & \text{if } x < 0 \end{cases}$$

## 9.6 Logarithmic differentiation

Logarithms can transform

- powers into products
- products into sums
- quotients into differences

Example

Find  $\frac{dy}{dx}$  for  $y = \left( \frac{(3x^2+4)(x+2)}{x^3+5x} \right)^{\frac{3}{5}}$

Take  $\ln$  of both sides,

$$\begin{aligned} \ln y &= \frac{3}{5} \ln \left( \frac{(3x^2+4)(x+2)}{x^3+5x} \right) \\ &= [\ln(3x^2+4) + \ln(x+2) - \ln(x^3+5x)] \end{aligned}$$

Differentiate both sides,

$$\begin{aligned} \frac{1}{y} \frac{dy}{dx} &= \frac{3}{5} \left( \frac{6x}{3x^2+4} + \frac{1}{x+2} - \frac{3x^2+5}{x^3+5x} \right) \\ \frac{dy}{dx} &= \frac{3}{5} \left( \frac{(3x^2+4)(x+2)}{x^3+5x} \right)^{\frac{3}{5}} \left( \frac{6x}{3x^2+4} + \frac{1}{x+2} - \frac{3x^2+5}{x^3+5x} \right) \end{aligned}$$

## 9.7 Indeterminate forms with powers

The limit  $\lim_{x \rightarrow 0^+} x^{\sin x}$  is of the indeterminate form  $0^0$ . The  $\ln$  function is continuous, so

$$\ln \left( \lim_{x \rightarrow 0^+} x^{\sin x} \right) = \lim_{x \rightarrow 0^+} \ln(x^{\sin x}) = \lim_{x \rightarrow 0^+} \sin x \ln x.$$

Recalling the pinching theorem,

$$0 \leq \lim_{x \rightarrow 0^+} |\sin x \ln x| \leq \lim_{x \rightarrow 0^+} |x \ln x| = 0 \implies \ln \left( \lim_{x \rightarrow 0^+} x^{\sin x} \right) = 0$$

## Chapter 10

# Hyperbolic Functions

### 10.1 Hyperbolic sine and cosine functions

**Definition:**

- The hyperbolic cosine function  $\cosh : \mathbb{R} \rightarrow \mathbb{R}$  is defined by

$$\cosh x = \frac{1}{2}(e^x + e^{-x}), \quad \forall x \in \mathbb{R}$$

- The hyperbolic sine function  $\sinh : \mathbb{R} \rightarrow \mathbb{R}$  is defined by

$$\sinh x = \frac{1}{2}(e^x - e^{-x}), \quad \forall x \in \mathbb{R}$$

Note:

$$\frac{d}{dx}(\sinh x) = \cosh x, \quad \frac{d}{dx}(\cosh x) = \sinh x,$$

Properties of  $\cosh$

- $\cosh$  is an even function
- $\cosh = 1, \cosh x \geq 1$  for all  $x \in \mathbb{R}$
- $\cosh x$  gets arbitrarily close to  $\frac{1}{2}e^{\pm x}$  as  $x \rightarrow \pm\infty$

Properties of  $\sinh$

- $\sinh$  is an odd function
- $\cosh x$  gets arbitrarily close to  $\pm \frac{1}{2}e^{\pm x}$  as  $x \rightarrow \pm\infty$

If  $x \in \mathbb{R}$ , then  $\cosh^2 x - \sinh^2 x = 1$ .

$$\begin{aligned} \cosh^2 x - \sinh^2 x &= \left( \frac{e^x + e^{-x}}{2} \right)^2 - \left( \frac{e^x - e^{-x}}{2} \right)^2 \\ &= \frac{1}{4} [e^{2x} + 2 + e^{-x} - (e^{2x} - 2 + e^{-2x})] \\ &= 1 \end{aligned}$$

## 10.2 Other hyperbolic functions

$$\begin{aligned}\tanh x &= \frac{\sinh x}{\cosh x}, & \coth x &= \frac{\cosh x}{\sinh x}, \\ \operatorname{sech} x &= \frac{1}{\cosh}, & \operatorname{cosech} x &= \frac{1}{\sinh}.\end{aligned}$$

## 10.3 Hyperbolic properties

### Difference of squares

$$\begin{aligned}\cosh^2 x - \sinh^2 x &= 1 \\ 1 - \tanh^2 x &= \operatorname{sech}^2 x \\ \coth^2 x - 1 &= \operatorname{cosech}^2 x\end{aligned}$$

### Sum and difference

$$\begin{aligned}\sinh(x \pm y) &= \sinh x \cosh y \pm \cosh x \sinh y \\ \cosh(x \pm y) &= \cosh x \cosh y \pm \sinh x \sinh y \\ \tanh(x \pm y) &= \frac{\tanh x \pm \tanh y}{1 \pm \tanh x \tanh y}\end{aligned}$$

### Double-angles

$$\begin{aligned}\sinh(2x) &= 2 \sinh x \cosh x \\ \cosh(2x) &= \cosh^2 x + \sinh^2 x \\ \tanh(2x) &= \frac{2 \tanh x}{1 + \tanh^2 x}\end{aligned}$$

## 10.4 Hyperbolic derivatives and integrals

### Derivatives

i am lazy.

## 10.5 The inverse hyperbolic functions

$\sinh : \mathbb{R} \rightarrow \mathbb{R}$  and  $\tanh : \mathbb{R} \rightarrow \mathbb{R}$  are increasing, so they are invertible.  $\cosh$  restricted to the domain  $[0, \infty)$  is invertible.

#### Example

Evaluate  $\sinh\left(\cosh^{-1}\frac{4}{3}\right)$ . Note that  $\cosh^{-1}(\cosh) = x$  for  $x \geq 0$  only.

$$\sinh^2 x = \cosh^2 x - 1 = \left(\frac{4}{3}\right)^2 - 1 = \frac{7}{9} \quad \Rightarrow \quad \sinh\left(\cosh^{-1}\frac{4}{3}\right) = \frac{\sqrt{7}}{3}.$$

## 10.6 Integration leading to the inverse hyperbolic functions

Omitted.

## **10.7 A summary of important hyperbolic formulae**

Omitted.

## **10.8 The catenary**

Omitted.